

What's in a parasitic gap?

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Abstract. Prior literature has debated whether multiple gaps in parasitic gap constructions have a shared antecedent or separate antecedents. I adduce novel arguments for a shared antecedent approach from connectivity effects in English: parasitic gaps obligatorily match the binding theoretic class and syntactic category of the filler. I develop an analysis of parasitic gaps that derives these facts by synthesizing two very different previous accounts—Nissenbaum 2000 and Nunes 2004—in a way that preserves key insights of both. My analysis also clarifies the source of ‘weakest crossover’ effects with parasitic gaps, establishes novel tests for identifying null operators, and paves the way for a novel account of the A/ $\bar{A}$ -distinction in parasitic gap licensing.

Keywords. parasitic gap; connectivity; category matching; Binding Theory; layering; parallel Merge; crossover; A/ $\bar{A}$ -distinction

1 Introduction

In parasitic gap (*pg*) sentences like (1), a single filler seems to bind more than one gap, neither of which c-commands the other.¹

- (1) $\overbrace{\text{It's the apple pie}_1}^{\text{filler}}$ $\overbrace{\text{that we should reheat } ___}_\text{(licensing) gap}$ $\overbrace{\text{[before we serve } pg_1 \text{ to the guests}]}$ parasitic gap

Since their discovery by Ross (1967:sec. 4.2.4.4), the following two questions have preoccupied prior work on parasitic gaps (see Culicover 2001:esp. sec. 1.3.2 for overview).

- (2) *The movement question*
Are parasitic gaps created by $\bar{A}$ -movement within the *pg*-containing constituent?
- (3) *The antecedent question*
If parasitic gaps are created by ($\bar{A}$ -)movement, what are their antecedents?

In this paper, I develop a novel analysis of parasitic gaps which synthesizes two very different approaches from the previous literature (Nissenbaum 2000, Nunes 2004) and which answers the questions in (2)–(3) in the following way: Parasitic gaps and their licensors are created by $\bar{A}$ -movement of a single, shared antecedent—the (overt) filler. On my account, (1) is derived as in (4), where the filler *the apple pie* initially occupies both gap positions (i.e., as an argument both of *reheat* and of *serve*), then moves to the edge of the adjunct island containing the parasitic gap and to the edge of the main clause vP, and ultimately moves to its left-peripheral surface position in the main clause (see Hornstein 2001:75–82 for a related proposal).

- (4) It's the apple pie₁ that we should [_{vP} ⟨the apple pie₁⟩ [_{vP} reheat ⟨the apple pie₁⟩]
[_{Adjunct} ⟨the apple pie₁⟩ before we serve ⟨the apple pie₁⟩ to the guests]].

I adduce evidence for each aspect of the analysis.

First, in line with earlier work, I argue that parasitic gaps are created by $\bar{A}$ -movement to the edge of a constituent containing them, often an island (section 2). Support for this conclusion

¹This is known as the *anti-c-command condition* (e.g., Taraldsen 1981:493, Chomsky 1982:40, (56), Engdahl 1983:22, (64), Haegeman 1984, Safir 1987, Ershova 2021, Hewett 2025d; see Grosu 1980:19, (26) for an early, related observation). Bruening and Al Khalaf (2017:24, fn. 12) identify a similar constraint on ATB-movement.

comes from various tests for $\bar{A}$ -movement inside the *pg*-containing constituent, including island sensitivity, parasitic gap licensing, and complementizer-trace effects. I also demonstrate, contra Lasnik and Stowell 1991, that parasitic gaps trigger all types of crossover, in line with their being formed by $\bar{A}$ -movement. I adduce new arguments that so-called “weakest crossover” effects with parasitic gaps obtain just in case the crossed pronoun can be (A-)bound in the relevant configuration in the absence of $\bar{A}$ -movement; in this, I build upon proposals by Hornstein (1995), Ruys (2004), and Safir (2004, 2019), among others. The evidence therefore militates against accounts in which no syntactic movement is involved in the creation of parasitic gaps (*pace*, e.g., Taraldsen 1981, Chomsky 1982, Haik 1985, Cinque 1990, Frampton 1990, Niinuma 2010:167–168) and against accounts which countenance movement, though not internal to the *pg*-containing XP (e.g., Nunes 2004, Agbayani and Ochi 2023).

Second, I marshal novel support for a shared antecedent analysis of parasitic gaps from two kinds of connectivity effects: Parasitic gaps and their licensors must match the filler in binding theoretic class and syntactic category (section 3). The shared antecedent analysis in (4) accounts for this connectivity by positing a syntactic representation of the filler in all gap positions. By contrast, I argue that connectivity is not straightforwardly explained by separate antecedent analyses which derive parasitic gaps via movement of a null operator. As I show, true null operators (e.g., in *tough*-constructions and restrictive relative clauses) behave differently, motivating a null operator-less treatment of parasitic gaps.

In section 4, I develop a particular version of the shared antecedent analysis according to which a QP shell is layered over a constituent undergoing $\bar{A}$ -movement (following ideas in Cable 2010b, Safir 2019, and Hewett 2025a). My analysis derives the aforementioned connectivity effects and sheds light on the nature of internal Merge. Then, in section 5, I argue that my analysis affords a novel syntactic account of the fact that only $\bar{A}$ -movement licenses parasitic gaps—in a nutshell, this fact is a manifestation of category connectivity under movement. In section 6, I summarize my contributions and consider future directions.

Before beginning, I will also note that, although I focus almost exclusively on cleft sentences, all the generalizations reported here hold for equivalent sentences using topicalization.

2 Parasitic gaps are created by $\bar{A}$ -movement

Parasitic gaps are gaps that are only grammatical in the presence of another gap (called the *licensing* gap) created by $\bar{A}$ -movement (Engdahl 1983). $\bar{A}$ -movement in the cleft sentence in (5a) licenses the parasitic gap in the adjunct clause headed by *before* (though note that this parasitic gap is optional and can be replaced by a coindexed pronoun, as in (5b)).

- (5) a. It's the apple pie₁ that we should reheat ____₁ [before we serve pg₁ to the guests].
 b. It's the apple pie₁ that we should reheat ____₁ [before we serve it to the guests].

Eliminating the licensing gap yields unacceptability.

- (6) * We should reheat the apple pie₁ [before we serve pg₁ to the guests].
 (7) * It's the apple pie₁ that we should reheat the tea [before we serve pg₁ to the guests].

In (6), there is no licensing $\bar{A}$ -dependency whatsoever, hence a parasitic gap is ungrammatical. In (7), the only $\bar{A}$ -dependency is the one connecting the filler to the parasitic gap, and that dependency illegally crosses an adjunct island boundary.

There are various indications that parasitic gap sentences involve movement of a constituent α from the parasitic gap to the edge of a constituent containing it (see, e.g., Grosu 1980:17ff., Contreras 1984, Chomsky 1986, Browning 1987, Munn 1992, 1994, 2001, Nissenbaum 1998, 1999, 2000, Nissenbaum and Schwarz 2011, Davis 2020, Arregi and Murphy 2022, 2024). I represent this analysis schematically in (8).

- (8) It's the apple pie₁ that we should reheat ____₁ [α_1 before we serve pg₁ to the guests].

In the remainder of this section, I summarize the evidence supporting this analysis, which

comes from island sensitivity (section 2.1), parasitic gap licensing (section 2.2), complementizer-trace effects (section 2.3), and crossover (section 2.4). I return to the identity of α in section 3.

2.1 Islands

The first indication that parasitic gaps are formed by movement comes from island constraints. While parasitic gaps may be separated from their licensing gaps by an island boundary, as in (9), no additional island boundaries can intervene—an observation traditionally attributed to Kayne (1983) and Longobardi (1985) (see also Grosu 1980:18, (23) and Chomsky 1982:52, (70a)). Thus, the examples in (10), which place two islands on the path between the parasitic gap and licensing gap, are unacceptable.

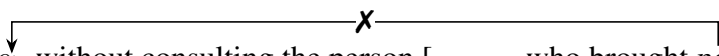
- (9) It's the shrimp₁ that they threw ____₁ away [Adjunct island because nobody wanted pg_1].
- (10) It's the shrimp₁ that they threw ____₁ away...
- a. ?? [Adjunct island without asking [Wh island why Joni wanted pg_1]].
 - b. * [Adjunct island without consulting the person [RC island who brought pg_1]].
 - c. * [Adjunct island without confirming the rumor [Complex NP island that we hated pg_1]].
 - d. * [Adjunct island after sulking [Adjunct island because nobody wanted pg_1]].
 - e. * [Adjunct island despite suspecting (that) [Subject island that we might want pg_1] was still possible].

Arregi and Murphy (2022:3, (8)) state this generalization as follows (see also Nissenbaum 2000:26):

- (11) *Island generalization*
At most one island boundary can intervene between a pg and its licensing gap.

The movement analysis of parasitic gaps straightforwardly predicts this effect. Movement of α to the edge of the island that is most local to the licensing gap must obey island constraints.

Example (12) illustrates for (10b).

- (12) ... [Adjunct island α_1 without consulting the person [RC island who brought pg_1]].
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- The diagram shows a horizontal line representing the movement path of α_1 . It starts at the position of the parasitic gap (indicated by a downward arrow) and ends at the position of the licensing gap (indicated by a downward arrow). An 'X' is placed above the line, indicating that this movement is blocked because it would cross the boundary of the RC island.

In this respect, movement of α parallels run-of-the-mill $\bar{A}$ -movement; compare the examples in (10) with those in (13).

- (13) a. ?? It was the shrimp₁ that they asked [_{Wh island} why Joni wanted ____₁].
 b. * It was the shrimp₁ that they consulted the person [_{RC island} who brought ____₁].
 c. * It was the shrimp₁ that they confirmed the rumor [_{Complex NP island} that we hated ____₁].
 d. * It was the shrimp₁ that they were sulking [_{Adjunct island} because nobody wanted ____₁].
 e. * It was the shrimp₁ that they suspected (that) [_{Subject island} that we might want ____₁] was still possible.

Importantly, movement to the edge of an island that does not cross that island does not incur any penalty, by hypothesis, explaining why a single island boundary can licitly separate parasitic gaps from their licensors (e.g., the adjunct island in (9)).

2.2 Parasitic gap licensing

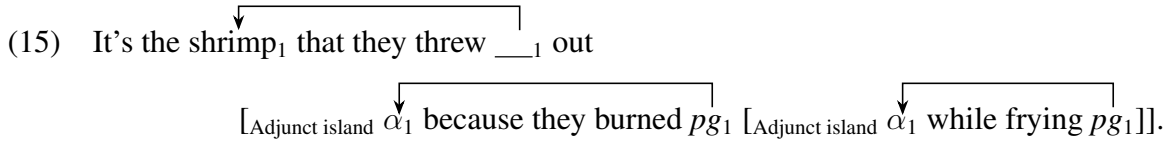
A second piece of evidence supporting the movement analysis of parasitic gaps comes from parasitic gap licensing itself. As the examples in (5)–(7) established, ($\bar{A}$ -)movement is necessary to license a parasitic gap.² Tellingly, then, one parasitic gap may license another (see, e.g., Ross 1967:191–192, Haik 1985:256, (33), Chomsky 1986:62, Cinque 1990:100, Williams 1990:270). Consider the following triple:

- (14) a. It's the shrimp₁ that they threw ____₁ out [_{Adjunct island} α_1 because they burned *pg*₁ [_{Adjunct island} while frying the tortillas]].
 b. * It's the shrimp₁ that they threw ____₁ out [_{Adjunct island} α_1 because they burned their fingers [_{Adjunct island} while frying *pg*₁]].

²See Hewett 2023:sec. 3.4 for arguments, contra Cinque 1990:110–115, that $\bar{A}$ -binding from Spec,CP is insufficient to license clause-internal parasitic gaps (e.g., in vP-level adjuncts). Note, however, that $\bar{A}$ -binding from a clause-internal position like Spec,vP, even in a base-generated (i.e., non-movement) $\bar{A}$ -dependency, may be sufficient, especially if we adopt an analysis of parasitic gap licensing like that in Nissenbaum 2000:esp. 99, (6) (see Hewett 2023:sec. 8.6.1 and Sichel 2023 for discussion). Because the distinction between movement through and base-generation in clause-internal positions is orthogonal to my main focus, I set this issue aside for the remainder of the paper.

- c. It's the shrimp₁ that they threw ____₁ out [Adjunct island α_1 because they burned pg_1 [Adjunct island while frying pg_1]].

$\bar{A}$ -movement in the main clause can license a parasitic gap in the outer adjunct headed by *because* ((14a)), though not in the inner adjunct headed by *while* ((14b)); this is in accordance with the island generalization in (11). Crucially, however, a parasitic gap in the *because* adjunct *can* license a parasitic gap in the *while* adjunct ((14c)). This is explained if the parasitic gap in the outer island is created by $\bar{A}$ -movement of α , as shown in (15).



2.3 Complementizer-trace effects

A third argument for *pg*-creating movement can be marshaled from complementizer-trace effects. As is well known, long-distance $\bar{A}$ -extraction shows a subject/non-subject asymmetry in many languages (see Pesetsky 2017 for overview). For instance, in English, extraction of a non-subject out of a finite declarative embedded clause optionally permits that clause to be introduced by the complementizer *that* ((16a)). However, as originally observed by Perlmutter (1968), extraction of a subject requires the complementizer most local to the extraction site to be null ((16b)).³ This particular interaction is known as the *that*-trace effect.

- (16) a. What₁ did he say [_{CP} { $\emptyset$ / that} Laura hid ____₁]?
 b. Who₁ did he say [_{CP} { $\emptyset$ / *that} ____₁ hid the rutabaga]?
 (Perlmutter 1968:214, (34a), (35))

Virtually all accounts of complementizer-trace effects derive them from movement: Either $\bar{A}$ -movement from a position to the immediate right of the complementizer is special, or $\bar{A}$ -movement from a position just below the complementizer is special (Pesetsky 2017:sec. 5).

³There is well known idiolectal variation in this domain (Pesetsky 2017:sec. 6.3). I, for one, find examples like (16b) with overt *that* to be perfectly acceptable. Needless to say, complementizer-trace effects can only be used to diagnose *pg*-creating movement in idiolects that detect a contrast in pairs like (16).

Hence, I assume that complementizer-trace effects are diagnostic of movement (see also Kenstowicz 1989).

In light of this diagnostic, consider Bresnan’s (1977:182) observation that parasitic gaps, like regular gaps, induce complementizer-trace effects. When a parasitic gap occupies the subject position of a finite declarative embedded clause, it cannot be immediately preceded by the complementizer *that* (see also Munn 1992:22, (49a) and Davis 2025:118).⁴

- (17) a. It’s the shrimp₁ that they threw ___₁ away [α_1 after discovering [_{CP} *pg*₁ had gone bad]].
 b. *It’s the shrimp₁ that they threw ___₁ away [α_1 after discovering [_{CP} *that pg*₁ had gone bad]].

Assuming that complementizer-trace effects diagnose movement, the *that*-trace effect in (17b) provides further evidence that parasitic gaps are created by movement.

2.4 Crossover

The fourth argument for *pg*-creating movement comes from crossover effects.⁵ The following primer on crossover effects closely follows the one presented in Hewett 2025a:sec. 3.1.5.

Crossover (Postal 1971) describes the failure of a quantificational phrase *Q* in an $\bar{A}$ -

⁴While speakers vary in the degree to which they accept examples like (17a) with embedded subject parasitic gaps (e.g., Andy Murphy (p.c.) finds such examples degraded), there remains a sharp contrast between (17a) and (17b). Note, further, that the (relative) acceptability of parasitic gap dependencies spanning a finite clause boundary, as in (17a), supports the claim that *pg*-creating movement is $\bar{A}$ -movement, since only $\bar{A}$ -movement in English crosses finite CPs (i.e., English lacks hyperraising; for recent discussion, see Zyman 2023b:sec. 3.2).

⁵Hewett (2023:ch. 7, 2025b) actually argues that crossover does not unambiguously diagnose movement based on his discovery that crossover effects are also manifest in base-generated resumptive $\bar{A}$ -dependencies. Following proposals in Büring 2004, Hewett accounts for crossover effects via constraints on $\bar{A}$ -binding (see also Hewett 2025a). Granting Hewett’s arguments, we can still use crossover to diagnose the presence of an $\bar{A}$ -dependency terminating in the parasitic gap, as I do in the main text.

position to bind a pronoun (which I call the *crossed pronoun*) distinct from the variable that Q binds, typically Q 's trace. Crossover is *strong* if the crossed pronoun c-commands the $\bar{A}$ -bound variable and it is *weak* if neither the crossed pronoun nor the $\bar{A}$ -bound variable c-commands the other (see, e.g., Wasow 1972, Lasnik and Funakoshi 2017, Safir 2017).⁶ Additionally, we can distinguish between *primary* crossover, where the crossed pronoun and $\bar{A}$ -bound variable are coindexed, and *secondary* crossover, where the crossed pronoun and the $\bar{A}$ -bound variable are contraindexed as a consequence of pied-piping (see, e.g., Postal 1993b, Safir 1996, Keine and Bhatt 2023, Hewett 2025b, and the references cited in Hewett 2023:331). The following English cleft sentences illustrate each of the four types of crossover:⁷

- (18) *Primary strong crossover*
It was some other guy₁ that he_{3/*1} told me I look like ____₁.
- (19) *Primary weak crossover*
It was some other guy₁ that [his_{3/*1} friends] told me I look like ____₁.
- (20) *Secondary strong crossover*
It was [some guy₁'s dog]₂ that he_{3/*1} told me I look like ____₂.
- (21) *Secondary weak crossover*
It was [some guy₁'s dog]₂ that [his_{3/*1} friends] told me I look like ____₂.

In all cases, the quantificational phrase *some (other) guy*, which is reflexively contained in a constituent in an $\bar{A}$ -position (the cleft pivot), fails to bind a pronoun (*him* or *his*) that is not bound by the clefted constituent's trace. In (18), the crossed pronoun c-commands the $\bar{A}$ -bound trace and is coindexed with it; this is primary strong crossover. In (19), by contrast, the crossed pronoun does not c-command the trace that it is coindexed with, hence this is primary weak crossover. In (20) and (21), the crossed pronoun and $\bar{A}$ -bound trace are not coindexed; rather, the trace is coindexed with a phrase pied-piped by the clefted quantificational phrase.

⁶Many speakers report weak crossover violations to be less severe than strong crossover violations (Wasow 1972). I set aside this dimension of variation for the remainder of the paper.

⁷As Postal (1993b:541–542) observes, while ‘simple’ quantificational expressions like *someone* and *everyone* cannot be clefted (**It was someone₁/everyone₁ that you told me I look like ____₁*), quantificational phrases that are restrictively modified (e.g., by adjectives, exceptive phrases, or relative clauses) can be.

This yields secondary crossover—strong in (20) because the crossed pronoun c-commands the trace and weak in (21) because it does not.⁸

We are now in a position to investigate crossover with parasitic gaps. As the examples in

⁸As mentioned at the end of section 1, similar facts hold for topicalization. Recent claims by Bruening (2022) and Chierchia (2024) that (PP) topicalization in English fails to trigger secondary weak crossover are false (just as Lasnik and Stowell (1991)’s claim that DP topicalization fails to trigger primary weak crossover is false, see Postal 1993b) and the empirical patterns that they allege substantiate such claims are confounded. As previous work has shown, topicalization of an argument DP properly containing a quantificational phrase induces secondary weak crossover effects (Safir 1996:323, fn. 7, 326, 1999:603, (42b) and Ruys 2004:137). Though not previously recognized, this includes conjunctions of DPs, one of which is a quantificational phrase.

- (i) [_{&P} Zendaya, the Olsen twins, and [exactly one man from Toronto]₁]₂, [his_{3/*1} parents] think you saw ____₂ at a party in LA.

Likewise, topicalization of argument PPs triggers secondary weak crossover.

- (ii) (Krishna₃’s friends rely on him for money...)
[_{PP} On every other guy₁’s dog]₂, (Joni thinks) [his_{3/*1} friends] rely ____₂ for emotional support.

In fact, topicalizing constituents of various other categories, such as APs ((iii)), VPs/vPs ((iv)), CPs ((v)), and participles ((vi)), uniformly induces secondary weak crossover.

- (iii) [_{AP} Proud of every woman₁]₂, [her_{3/*1} parents] certainly (believe you) are ____₂.
(iv) [_{VP/vP} Speak to every woman₁]₂, [her_{3/*1} parents] (claim you) did ____₂.
(v) [_{CP} That exactly one woman₁ was under surveillance]₂, [her_{3/*1} parents] never realized ____₂.
(vi) [_{PartP} Sitting next to every woman₁]₂, her_{3/*1} parents insist ____₂ should be the mayor of the city.

Bruening (2022) and Chierchia (2024) base their claims on examples in which a locative adjunct PP properly containing a quantificational DP appears in the left periphery of the clause it modifies and secondary weak crossover effects are suspended. Note, however, that such examples are amenable to an alternative, non-movement parse: The locative adjunct PP can be base-generated in a left-peripheral position (e.g., its surface position), with the consequence that no crossover effect is expected. Strong support for this alternative comes from the novel observation that, in *long-distance* topicalization of adjunct PPs, secondary weak crossover reappears when the dependency crosses a pronoun one or more clauses above where the adjunct PP is interpreted, as in (vii).

- (vii) [At exactly one guy₁’s defense]₂, [his_{3/*1} advisor] believes [_{CP} we burst into tears ____₂].

(22)–(25) show, all four types of crossover are observable in parasitic gap dependencies.⁹

(22) *Primary strong crossover*

It was some other guy₁ that, [α_1 after he_{3/?*1} told me I look like pg_1], I said ____₁ should take a photo of me.

(23) *Primary weak crossover*

It was some other guy₁ that, [α_1 after [his_{3/?*1} friends] told me I look like pg_1], I said ____₁ should take a photo of me.

(24) *Secondary strong crossover*

It was [some guy₁'s dog]₂ that, [α_2 after he_{3/?*1} told me I look like pg_2], I took a photo with ____₂.

(25) *Secondary weak crossover*

It was [some guy₁'s dog]₂ that, [α_2 after his_{3/?*1} friends told me I look like pg_2], I took a photo with ____₂.

Consider first (22): The parasitic gap licensed by $\bar{A}$ -extraction of the quantificational phrase *some other guy* cannot be bound by *he*. This indicates that parasitic gaps trigger primary strong crossover (see Barss 1986:481; Cinque 1990:150; Lasnik and Stowell 1991:709, Postal 1993a:752, fn. 14, (ivb), 2001:246, n. 11, Kennedy 1997:702–704, Culicover 2001:33, (89b), and Safir 2004:84, (54b)). Similarly, in (23), coindexation between the pronoun *his* and the parasitic gap which it neither binds nor is bound by is significantly degraded, reflecting primary weak crossover (Hornstein 1995:105–106, Levine and Hukari 2006:317–318, esp. (26),

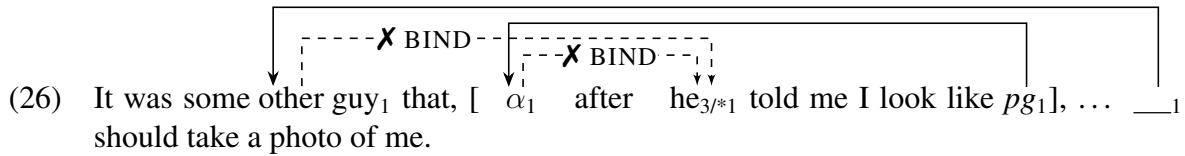
To derive a reading where the adjunct PP modifies the embedded clause, that PP must undergo $\bar{A}$ -movement from within the embedded CP to its surface position. This movement necessarily crosses the pronoun *his* in the matrix clause and, as expected, this triggers a secondary weak crossover effect. Finally, note that adjunct PPs which are unambiguously generated low and on the right, such as passive *by*-phrases, trigger secondary weak crossover even in monoclausal topicalization ((viii)), reinforcing the conclusion that topicalization induces crossover effects like normal.

(viii) [By exactly one guy₁'s car]₂ [his_{3/?*1} friends] were hit ____₂.

⁹Examples (22)–(25) use subject $\bar{A}$ -extraction to license a parasitic gap in a preposed adjunct, a possibility recognized by Haegeman (1984) and Chomsky (1986:54, citing observations by Giuseppe Longobardi in unpublished work); see also Davis 2025:118, (28) and Hewett 2025d:app. B.

(30), and Al Khalaf 2015:171, (379a); see Bruening and Al Khalaf 2017:sec. 2.3 for related discussion). Finally, in (24) and (25), there can be no coindexation between the pied-piping quantificational expression *some guy* and the crossed pronoun (*he/his*) inside the *pg*-containing adjunct. These examples illustrate secondary strong and weak crossover with parasitic gaps, respectively (on secondary strong crossover with parasitic gaps, see Barss 1986:472, Williams 1986:285, (54), 1990:271–272, esp. (33), and Bruening and Al Khalaf 2017:16–18; on secondary weak crossover with parasitic gaps, see Levine and Hukari 2006:316, (24)).

The movement analysis of parasitic gaps predicts these effects without further ado. $\bar{A}$ -movement of α from the *pg* to the edge of the containing island cannot feed variable binding of a pronoun from α 's landing site, since pronominal variable binding is only possible from A-positions (Reinhart 1983, Buring 2004). Similarly, since the filler occupies an $\bar{A}$ -position at the edge of the matrix clause, it too will be unable to bind a pronoun crossed by movement of α .¹⁰ Without an available binder, the crossed pronoun cannot be interpreted as a bound variable. Example (26) illustrates for the primary strong crossover configuration in (22).



The other three types of crossover with parasitic gaps are similarly accounted for.

Despite this, as famously observed by Lasnik and Stowell (1991), primary weak crossover effects are unexpectedly missing with certain parasitic gaps. Consider (27).

- (27) It was some other girl₁ that they gossiped about ₁ [α_1 despite her₁ teacher's having vouched for *pg*₁].
(adapted from Lasnik and Stowell 1991:694, (23a))

Coindexation between the parasitic gap and the crossed pronoun *her* in the adjunct island is

¹⁰See Buring 2004 and Hewett 2023:ch. 7, 2025b on how secondary crossover can be derived by prohibiting indirect binding (i.e., binding in the apparent absence of c-command) by constituents in $\bar{A}$ -positions (and see Hewett 2025a:sec. 5.5 for a related, but distinct, account).

perfectly acceptable—an effect Lasnik and Stowell dub *weakest crossover* (see Huybregts and Riemsdijk 1985:186, n. 13, (i) for a similar observation in Dutch and see Barss 1986:499, fn. 19, (3) on the suspension of *secondary* weak crossover effects with English parasitic gaps). This observation has led much subsequent work to conclude that parasitic gaps *never* trigger (primary) weak crossover, contrary to the evidence in (23). If correct, that conclusion would potentially challenge the movement analysis of parasitic gaps I have developed in this section.

Fortunately, the puzzle of weakest crossover with parasitic gaps relents to a simple solution. As suggested by Hornstein (1995:101–106), we can explain the absence of weak crossover in examples like (27) once we recognize that there *is* a potential binder for the crossed pronoun that is not in an $\bar{A}$ -position—namely, the licensing gap in the main clause (see Ruys 2004 and Safir 2004:sec. 3.6, 2019:316–317 for related proposals). On such an account, (27) is predicted to be acceptable for the same reason that (28) is, where the latter crucially involves variable binding into the adjunct clause *without* $\bar{A}$ -movement.¹¹

(28) They gossiped about some other girl₁ [despite her₁ teacher’s having vouched for her₁].

Thus, weak crossover is predicted to reemerge with parasitic gaps if the crossed pronoun cannot be bound by the licensing gap. (23) is one such environment.

(29) [After his_{3/*1} friends told me I look like him_{3/*1}], I said some other guy₁ should take a photo of me.

The quantificational subject *some other guy* cannot bind pronouns in the preposed adjunct to its left, so neither can the licensing gap in (23) (see also Barss 1986:497, (260b)).¹²

¹¹In order to reconcile the fact that *some other girl* (or its trace) can bind a variable to its right in a clause-final adjunct with the anti-c-command condition on parasitic gaps (on which see footnote 1), we may need to abandon a c-command–based definition of binding in favor of something more coarse-grained. See Hewett 2025d:sec. 5.1, app. B for an analysis of left-to-right binding in English and the anti-c-command condition that is compatible with the facts presented here. For other work recognizing a tension between binding and the anti-c-command condition, see Contreras 1984 (and the response in Safir 1987) and Arregi and Murphy 2022:33–35, esp. fn. 16.

¹²Weak crossover effects are detectable in at least two other *pg*-licensing configurations where the licens-

Two novel lines of evidence support the idea that weakest crossover with parasitic gaps is the result of exceptional A-binding of the crossed pronoun by the licensing gap. First, the crossed pronoun can be a bound anaphor, such as *each other*.

- (30) It was [two girls from a nearby town]₁ that he gossiped about ____₁ [α_1 despite each other₁'s teachers having vouched for *pg*₁].

Anaphors must be A-bound (Reinhart 1983), as in (31); compare (32), where topicalization—a type of $\bar{A}$ -movement that also displays weakest crossover according to Lasnik and Stowell (1991)—fails to feed anaphor binding (see also Charnavel and Sportiche 2016:68, (68a)).

- (31) He gossiped about [two girls from a nearby town]₁ despite each other₁'s teachers having vouched for them₁.
 (32) * [Two girls from a nearby town]₁, each other₁'s teachers said you vouched for ____₁.

In light of this, I submit that *each other* in (30) must be A-bound by the licensing gap and not bound by a constituent in an $\bar{A}$ -position (e.g., α or the filler).

The second indication that A-binding underlies weakest crossover with parasitic gaps comes from examining what Hewett (2023:450–451, fn. 117) calls ‘joint binding.’ Hewett observes that multiple binders can simultaneously bind a single pronoun and he represents such joint binding with a conjunction of indices on the bound variable (i.e., ‘1+2’ in (33)).

- (33) No professor₁ told any student₂ that they₁₊₂ were meeting at six.
 (Hewett 2023:450, fn. 117, (i))

ing gap cannot A-bind the crossed pronoun. The first involves the *pg*-containing constituent attaching one or more clauses above the licensing gap in a rightward/clause-final position ((i)) and the second involves the *pg*-containing constituent being an argument that c-commands the licensing gap ((ii)).

- (i) a. It was some other girl₁ that I only thought [_{CP} you hired ____₁] [because her_{3/?*1} mom told me you hired *pg*₁].
 b. I only thought [_{CP} you hired some other girl₁] [because her_{3/*1} mom told me you hired her_{3/*1}].
 (ii) a. It was somebody else₁ that [their_{3/?*1} mom's pictures of *pg*₁] reminded Joni she needs to talk to ____₁.
 b. [Their_{3/*1} mom's pictures of them_{3/*1}] reminded Joni she needs to talk to somebody else₁.

Hewett also notes that all joint binders must be of the same type—that is, they must all be A-binders, as in (33), or $\bar{A}$ -binders (see Hewett 2023:450, fn. 117 for an example and pertinent discussion). Mixing A- and $\bar{A}$ -binders, as in (34), is not possible.¹³

- (34) * Which professor₁ were you content despite the fact that we didn't suggest to any student₂ that they₁₊₂ were meeting at six? (Hewett 2023:451, fn. 117, (iii))

Although this restriction remains unaccounted for, to my knowledge, we can nonetheless leverage it to probe the source of weakest crossover with parasitic gaps. Consider (35).

- (35) Which professor₁ did every student₂ meet with ___₁ [α_1 only after their₁₊₂ mutual friend spoke positively about *pg*₁]?

Pg-creating $\bar{A}$ -movement of α within the adjunct does not trigger weak crossover vis-à-vis the crossed pronoun *their*: This is a weakest crossover effect. Importantly, in order to derive the jointly bound reading of *their* (i.e., with index '1+2'), both of this pronoun's binders must occupy the same type of position. Since the matrix subject *every director*, which bears index '2,' occupies an A-position, its joint-binding counterpart bearing index '1' must also occupy an A-position. The only potential binder meeting those conditions is the licensing gap in the main clause (contrast *which actor* and α , which occupy $\bar{A}$ -positions). If we eliminate this A-binder as in (36), joint binding is impossible due to the constraint on uniform binding types, and weak crossover resurfaces (see Sportiche 1983:179 for a related observation).

- (36) * Which professor₁ did every student₂ hear their₁₊₂ mutual friend speak positively about ___₁?

In light of these arguments, I conclude that weakest crossover fails to motivate a special treatment of parasitic gaps and their binders vis-à-vis other $\bar{A}$ -bound gaps, contra Lasnik and Stowell 1991 (which treats parasitic gaps as null, $\bar{A}$ -bound epithets) and much subsequent work. Rather, many parasitic gap configurations simply make available an additional A-binder

¹³Because resumptive pronouns are generally most acceptable in English inside islands, the bound variable *they* in (34) is positioned in an adjunct island to give the example its best shot at acceptability.

for the crossed pronoun in the form of the licensing gap. When this possibility is controlled for, weak crossover emerges as expected (see (23) and (25)). The upshot is that parasitic gaps, like other gaps created by $\bar{A}$ -movement, trigger the full suite of crossover effects.

To summarize this section, I have adduced four separate arguments that parasitic gaps are created by $\bar{A}$ -movement of an element α to the edge of a constituent containing the parasitic gap (often an island), as illustrated in (37).

(37) It's the pie_1 that we should reheat $__1$ [α_1 before we serve pg_1 to the guests].

The arguments were based on island-sensitivity, parasitic gap licensing, complementizer-trace effects, and crossover.¹⁴ These findings therefore militate against analyses of parasitic gaps as in situ empty pronominals (*pace* Taraldsen 1981, Chomsky 1982, Haik 1985, Cinque 1990, Frampton 1990, Niinuma 2010:167–168) and against accounts which do not posit movement internal to the *pg*-containing constituent (e.g., Nunes 2004, Agbayani and Ochi 2023). What remains to be determined is the nature and identity of α . This is the focus of the next section.

3 Connectivity effects with parasitic gaps: Arguments for a shared antecedent approach

Prior work has debated the identity of parasitic gap antecedents. In other words: What is α in (37)? Following Nissenbaum (2000:24–29), two classes of approach can be distinguished.

Shared antecedent analyses propose that the licensing gap and parasitic gap are created by movement of a single constituent—the filler (e.g., Sag 1983, Williams 1990, Munn 1994, Hornstein 2001, Levine et al. 2001, Hornstein and Nunes 2002, Nunes 2004, Bruening and Al Khalaf 2017). A schematic shared antecedent analysis of (37) that is compatible with the preceding evidence for *pg*-creating movement is given in (38).

¹⁴Parasitic gap dependencies also induce crossing effects (Frampton 1990:58, fn. 21). If such crossing effects diagnose movement (Pesetsky 1987b:104–106), then this would constitute an additional argument in favor of the movement analysis of parasitic gaps in (37).

(38) *A shared antecedent analysis of parasitic gaps*

It's the pie₁ that we should reheat ⟨the pie₁⟩ [⟨the pie₁⟩ before we serve ⟨the pie₁⟩ to the guests].

The filler *the pie* is initially generated as an argument both of the main clause verb *reheat* and of the adjunct clause verb *serve*. It then undergoes $\bar{A}$ -movement to the edge of the adjunct, and, ultimately, moves to its surface position at the left edge of the main clause.

By contrast, *separate antecedent* analyses argue that parasitic gaps are created by movement of an element distinct from the filler, usually a null operator *Op*, that terminates at (or near) the edge of the maximal island containing the *pg* (e.g., Contreras 1984, Clark 1985, Stowell 1985, Chomsky 1986, Browning 1987, Postal 1998, Nissenbaum 2000, Munn 2001, Davis 2020, Arregi and Murphy 2022, 2024).

(39) *A separate antecedent analysis of parasitic gaps*

It's the pie₁ that we should reheat ⟨the pie₁⟩ [*Op*₁ before we serve ⟨*Op*₁⟩ to the guests].

On such approaches, the filler and null operator are linked via binding, but, crucially, the filler never occupies a position internal to the *pg*-containing constituent.

In this section, I adduce two novel arguments for a shared antecedent approach based on connectivity effects: Parasitic gaps display connectivity with the filler in binding class (section 3.1) and syntactic category/lexical identity (section 3.2). I demonstrate that neither type of connectivity is predicted on a separate antecedent approach, since other, true null operator dependencies (e.g., in *tough*-constructions) behave differently from parasitic gaps. A major upshot of this discussion is that I identify two tests which diagnose the presence vs. absence of null operators.¹⁵

¹⁵A third type of connectivity often discussed in relation to parasitic gaps is *case-matching*; see, for instance, É. Kiss 1985, Tellier 1991:134–135, Franks 1992, Kathol 2001, Bondaruk 2003, and Himmelreich 2017. While basic case connectivity between fillers, licensing gaps, and parasitic gaps can be straightforwardly accounted for by shared antecedent approaches, it remains to be seen whether such approaches can also explain the ameliorating

3.1 Parasitic gaps display binding connectivity with their fillers

A compelling case for a movement analysis of filler-gap dependencies can be made from the fact that properties of the filler determine the distribution of its gap(s). As previous authors have shown, the Binding-Theoretic class of a filler—whether it is an R-expression/quantifier, a pronoun, or an anaphor—determines which Binding-Theoretic condition the gap will obey (e.g., Barss 1986:477ff., Williams 1986:288–289, Postal 2004:211–213, Safir 2004:81, Culicover 2013:132–133, 136–137). A gap whose filler is an R-expression/quantifier obeys Condition C/strong crossover ((42a)): It cannot be A-bound. A gap whose filler is a pronoun obeys Condition B ((42b)): It cannot be A-bound in its domain. And finally, a gap whose filler is an anaphor obeys Condition A ((42c)): It must be A-bound in its domain.

- (40) a. *Condition C/Strong crossover*
 It was some other girl₁ that she_{3/*1} taught you to describe ____₁.
 b. *Condition B*
 i. It was her₁ that she₁ taught you to describe ____₁.
 ii. It was her₁ that you taught her_{3/*1} to describe ____₁.
 c. *Condition A*
 It was herself₁ that you taught her_{1/*3} to describe ____₁.

We can account for these facts by positing a representation of the filler in the gap position along the lines of (41), as the lower representation of the filler will be subject to the Binding Conditions (see, e.g., the Copy Theory of Movement (Chomsky 1993) or multidominance (Citko 2005, Johnson 2012)). Thus, the acceptability of the examples in (40), which involve $\bar{A}$ -movement, parallels the acceptability of the examples in (42), which do not.

- (41) a. *Condition C/Strong crossover*
 It was some other girl₁ that she_{3/*1} taught you to describe ⟨some other girl₁⟩.
 b. *Condition B*
 i. It was her₁ that she₁ taught you to describe ⟨her₁⟩.
 ii. It was her₁ that you taught her_{3/*1} to describe ⟨her₁⟩.

effects of syncretism on erstwhile unacceptable case mismatches. I leave this as a task for future work.

- c. *Condition A*
It was herself₁ that you taught her_{1/*3} to describe ⟨herself₁⟩.
- (42) a. *Condition C/Strong crossover*
She_{3/*1} taught you to describe some other girl₁.
- b. *Condition B*
 - i. She₁ taught you to describe her₁.
 - ii. You taught her_{3/*1} to describe her₁.
- c. *Condition A*
You taught her_{1/*3} to describe herself₁.

Strikingly, as originally observed by Barss (1986:477ff.), parasitic gaps display the exact same type of binding connectivity with their fillers. A parasitic gap coindexed with a filler that is a quantificational phrase cannot be A-bound ((43)), a parasitic gap coindexed with a pronominal filler cannot be locally A-bound ((44)), and a parasitic gap coindexed with an anaphoric filler can be locally A-bound ((45); see also Longobardi 1985:186, fn. 23, Levine et al. 2001:211, (56), Al Khalaf 2015:174, (385), and Bruening and Al Khalaf 2017:2–3, 7–12, but see Postal 2001:224–225 for a different judgment).

- (43) *Condition C/Strong crossover*
It was some other girl₁ that I recognized ____₁ [α_1 because she_{3/*1} taught you to describe *pg*₁ so well].
- (44) *Condition B*
 - a. It was her₁ that I recognized ____₁ [α_1 because she₁ taught you to describe *pg*₁ so well].
 - b. It was her₁ that I recognized ____₁ [α_1 because you taught her_{3/*1} to describe *pg*₁ so well].
- (45) *Condition A*
It was herself₁ that Joni₁ described ____₁ [α_1 after PRO₁ seeing *pg*₁ in the mirror].

Eliminating the parasitic gap renders the deviant examples acceptable.

- (46) *No Condition C/strong crossover effect without a parasitic gap* (cf. (43))
It was some other girl₁ that I recognized ____₁ [because she₁ taught you to describe her₁ so well].
- (47) *No Condition B effect without a parasitic gap* (cf. (44b))
It was her₁ that I recognized ____₁ [because you taught her₁ to describe herself₁ so well].

This indicates that binding of the parasitic gap is what determines the (un)acceptability of examples (43)–(45).

A shared antecedent analysis of parasitic gaps straightforwardly explains this binding connectivity. On a shared antecedent approach, the licensing gap and the parasitic gap are gaps left by movement of the filler. Therefore, both gaps contain a representation of the filler subject to the Binding Theory. Accordingly, (43)–(45) would be analyzed as in (48)–(50), respectively.

- (48) It was some other girl₁ that I recognized ⟨some other girl₁⟩ [⟨some other girl₁⟩ because she_{2/*1} taught you to describe ⟨some other girl₁⟩ so well].
- (49) a. It was her₁ that I recognized ⟨her₁⟩ [⟨her₁⟩ because she₁ taught you to describe ⟨her₁⟩ so well].
b. It was her₁ that I recognized ⟨her₁⟩ [⟨her₁⟩ because you taught her_{2/*1} to describe ⟨her₁⟩ so well].
- (50) It was herself₁ that Joni₁ described ⟨herself₁⟩ [⟨herself₁⟩ after PRO₁ seeing ⟨herself₁⟩ in the mirror].

Binding connectivity with parasitic gaps reduces to a garden variety reconstruction effect, explained by the same machinery that accounts for reconstruction with non-parasitic gaps.

Consider now what a separate antecedent analysis of binding connectivity would look like. In all cases, the antecedent of the parasitic gap would be a null operator, as shown in (51)–(53).

- (51) *Condition C/strong crossover*
It was some other girl₁ that I recognized ⟨some other girl₁⟩ [*Op*₁ because she_{2/*1} taught you to describe ⟨*Op*₁⟩ so well].
- (52) *Condition B*
a. It was her₁ that I recognized ⟨her₁⟩ [*Op*₁ because she₁ taught you to describe ⟨*Op*₁⟩ so well].
b. It was her₁ that I recognized ⟨her₁⟩ [*Op*₁ because you taught her_{2/*1} to describe ⟨*Op*₁⟩ so well].
- (53) *Condition A*
It was herself₁ that Joni₁ described ⟨herself₁⟩ [*Op*₁ after PRO₁ seeing ⟨*Op*₁⟩ in the mirror].

We would therefore expect the parasitic gap to track the binding properties of *Op*, not those of the filler—at least not directly. This raises an important question for separate antecedent

analyses: Which of the Binding Conditions is *Op* subject to? There exist two answers to this question in the prior literature, neither of which accounts for all the facts.

First, Chomsky (1982:esp. 40–41) proposes that parasitic gaps, being $\bar{A}$ -bound, are variables. Assuming that variables function as R-expressions for the Binding Theory (e.g. Chomsky 1981), parasitic gaps ought to obey Condition C (see also Contreras 1993). Such an analysis would derive (51), but would leave the Condition B and A effects in (52) and (53), respectively, unaccounted for. Alternatively, various authors have proposed that *Op* is a null pronoun *pro* (e.g. Browning (1987), Postal (1998), Munn (2001), McCloskey (2002), and Hewett (2023:esp. 256–258)). The *pro* analysis is similarly too weak: It derives the Condition B effect in (52) but not the Condition C and A effects. The fact that parasitic gaps track the binding properties of the *filler* is mysterious for extant separate antecedent approaches.

To make a separate antecedent approach truly viable, then, we would need to invoke some mechanism to force *Op* and the filler to match in Binding-Theoretic class. Not only would this hypothetical mechanism be suspect given its stipulative nature, the mechanism would need to be conspicuously *unavailable* in other environments. For instance, the null operator in restrictive relative clauses ((54)) and in *tough*-constructions ((55)) uniformly triggers Condition C/strong crossover effects, regardless of whether *Op*'s antecedent (i.e., the relative head in relative clauses or the subject of the *tough* predicate) is an R-expression or a pronoun.

- (54) *Op uniformly triggers Condition C/strong crossover effects in restrictive relatives*
 a. {The girl₁ / She₁} thinks she₁ is a genius.
 b. [{She₁ / The girl₁} [_{CP} *Op*₁ that she_{3/*1} thinks ⟨*Op*₁⟩ is a genius]] is in fact a fool.
 c. [{She₁ / The girl₁} [_{CP} *Op*₁ that ⟨*Op*₁⟩ thinks she₁ is a genius]] is in fact a fool.
 (55) *Op uniformly triggers Condition C/strong crossover effects in tough-constructions*¹⁶
 a. It was tough to convince {the girl₁ / her₁} to let us accompany her₁.

¹⁶This finding militates against direct movement approaches to *tough*-constructions like those in Rosenbaum 1967:194–195, Bresnan 1971, Postal 1971, Postal and Ross 1971, Brody 1993, Hornstein 2001, Hartman 2011, 2012, Longenbaugh 2017, and Schwarzer 2019, a.o., according to which the *tough*-subject moves to its surface position from the gap site. If *she* moved from the gap in (55b), we would expect the gap to obey Condition B,

- b. {She₁ / Joni₁} was tough [_{CP} Op₁ to convince her_{3/*1} to let us accompany ⟨Op₁⟩].
- c. {She₁ / Joni₁} was tough [_{CP} Op₁ to convince ⟨Op₁⟩ to let us accompany her₁].

Examples (54a) and (55a) contain acceptable baseline sentences without extraction, where no Binding Condition is violated. Examples (54b) and (55b) show that $\bar{A}$ -movement of *Op* from the lower coindexed position induces a Condition C/strong crossover effect, regardless of whether *Op*'s antecedent is a pronoun or an R-expression.¹⁷ Note, as a control, that extracting *Op* from the higher coindexed position in (54c) and (55c) is perfectly licit because *Op* remains unbound within its scope. Thus, a matching null operator analysis of parasitic gaps fails to account for the idiosyncratic binding properties of null operators in parasitic gap constructions without additional (likely stipulative) assumptions.

I conclude that null operators and their traces qualify as R-expressions/quantifiers for the Binding Theory, *pace* Browning 1987, a.o. They are expected to always (and only) obey Condition C. As we saw above, parasitic gaps are different: They are contextually sensitive to the binding properties of the filler. Hence, binding connectivity provides strong evidence against a null operator analysis of parasitic gaps and in favor of a shared antecedent approach.¹⁸ Before not Condition C, contrary to fact; compare (41b). Hicks' (2003, 2009) smuggling analysis of *tough*-movement will face similar difficulties if the binding properties of the gap are determined by the smuggled phrase.

¹⁷For other work recognizing Condition C/strong crossover effects with the null operator in *tough*-constructions, see Chomsky 1981:204ff., Lasnik and Stowell 1991:709, (67a), Safir 2004:84, (54a), Levine and Hukari 2006:369, (81), and Rezac 2006:313 (and for an early related observation, see Postal 1976:218, (22c)). On Condition C/strong crossover effects in the related *tool/enough*-construction, see Cinque 1990:150, (165).

¹⁸Karlos Arregi (p.c.) and Luismi Toquero-Pérez (p.c.) wonder whether a 'matching' analysis of null operators could derive binding connectivity with parasitic gaps. In short, it could not. Matching analyses have been proposed for various other null operators, for instance in restrictive relative clauses (Sauerland 1998, Bhatt 2002, Salzmann 2017b), comparative deletion (Kennedy 2002), prolepsis (Salzmann 2017a,b, 2019), and *tough*-movement (Park 2022); see Hewett 2023:esp. ch. 6.2 for a 'matching'-like analysis of (resumptive) pronouns and their antecedents. On these analyses, the null operator is taken to be a null determiner whose NP comple-

turning to a second type of connectivity with parasitic gaps in section 3.2, I will make a brief excursus to address binding connectivity with internally complex fillers.

3.1.1 Excursus: Binding connectivity with internally complex fillers

Despite the clear evidence for binding connectivity between parasitic gaps and their fillers, reconstruction into parasitic gaps with internally complex fillers is more complex. Consider the contrast in (56), attributed to Kearney (1983) by Chomsky (1986:60) (see also Haïk 1985:284–292, Barss 1986:461–469, Munn 1992:9–13, and Citko 2005:493, (44)).

- (56) a. [Which pictures of himself₁]₂ did John₁ sell ____₂ [before Mary₃ had a chance to look at *pg*₂]?
 b. * [Which pictures of himself₁]₂ did Mary₃ sell ____₃ [before John₁ had a chance to look at *pg*₂]?

ment is deleted under identity with some antecedent (e.g. the relative head in a relative clause or the head of a comparative). Matching analyses therefore predict connectivity only with NP-internal material. However, as I have shown, D-related material also reconstructs to the parasitic gap position (e.g. quantificational determiners, see (43)). Hence, an NP-matching null operator analysis of parasitic gaps would fail on empirical grounds.

A defender of the matching analysis might therefore propose that it is the entire DP in the *pg*-containing constituent that matches the filler and deletes. This solution faces at least three challenges. First, a DP-matching analysis of (43) would be forced to posit a separate (identical) quantifier in the *pg*-containing clause, but this is clearly wrong, since (43) does not have the interpretation of (i).

- (i) It was some other girl₁ that I recognized ____₁ because she₃ taught you to describe some other girl so well.

Second, as examples (54)–(55) showed, other null operators do *not* match their antecedents in binding class. Without a way to restrict the DP-matching analysis to null operators in parasitic gap dependencies, this analysis thus overgenerates. Finally, one of the claimed advantages of matching analyses of null operators is that they can account for the *absence* of Condition C reconstruction in various contexts (see Salzmann 2019:sec. 2.2 for overview and references). However, as we have seen, parasitic gaps uniformly trigger Condition C/strong crossover effects. For these reasons, I reject matching null operator analyses of parasitic gaps.

(Nissenbaum 2000:30, (14))

Example (56a) shows that an anaphor which is properly contained inside the filler (e.g. *himself*) can reconstruct to the licensing gap position and be locally bound by an antecedent in the main clause (e.g. *John*). By contrast, the unacceptability of (56b) suggests that reconstruction solely to a parasitic gap inside a clause-final vP-level adjunct is not possible. Faced with this asymmetric reconstruction pattern, much previous literature has concluded, contrary to what I have been arguing, that there is no representation of the filler in the parasitic gap position. This is often taken to follow immediately on the null operator analysis (though see Browning 1987:162–165 for critical discussion).

There are at least two problems with this conclusion. First, null operators often *do* permit (anaphor) binding connectivity with internally complex fillers, as in *tough*-movement ((57)), *too/enough*-constructions ((58)), and comparatives ((59)) (see also Salzmann 2017b:301–303 on reconstruction for Condition A with null operators in prolepsis constructions).

- (57) [This side of herself₁]₂ was tough [*Op*₂ for John to get Mary₁ to deal with ____₂].
(slightly adapted from Pesetsky 2013:146, (43b); see also Browning 1987:164, (25a), Pesetsky 1987a, and Mulder and den Dikken 1992:310, fn. 8, (ii), citing David Lebeaux (p.c.); but see Poole et al. 2024 for critical discussion)
- (58) [Those pictures of himself₁]₂ are too embarrassing [*Op*₂ for John₁ to show us ____₂].
(Browning 1987:164, (25b))
- (59) More [pictures of himself₁]₂ than [*Op*₂ John₁ cared to think about ____₂] were handed out to his friends.
(Browning 1987:164, fn. 15, (ii))

This is predicted by ‘matching’ analyses of null operators (see footnote 18). Therefore, the lack of reconstruction to the parasitic gap position in (56b) does not actually follow on a null operator/separate antecedent analysis, at least not without additional assumptions.

The second problem is that, as observed by Munn (1994), Levine and Hukari (2006), and Bruening and Al Khalaf (2017), parasitic gaps *do* permit Condition A reconstruction with internally complex fillers when the parasitic gap linearly precedes the licensing gap. Consider the examples in (60).

- (60) a. [Which pictures of himself₁]₂ did John₁ put ____₂ into circulation [once Mary₃ finally approved of *pg*₂]?
 b. * [Which pictures of herself₁]₂ did John₁ put ____₂ into circulation [once Mary₃ finally approved of *pg*₂]?
 c. [Which pictures of herself₁]₂, [once Mary₃ finally approved of *pg*₂], did John₁ put ____₂ into circulation?
 (modeled after similar examples in Levine and Hukari 2006:49, (88))

Examples (60a) and (60b) replicate the contrast from (56): Reconstruction to the licensing gap is possible, whereas reconstruction to a rightward parasitic gap is not. Nevertheless, if the *pg*-containing adjunct is preposed to the left such that the parasitic gap precedes the licensing gap as in (60c), reconstruction to the parasitic gap suddenly becomes possible. Similarly, Munn (1994:407) provides the example in (61), illustrating that an anaphor properly contained inside the filler can be reconstructed to a parasitic gap in a subject that precedes the licensing gap (see also Al Khalaf 2015:173, (382); but cf. Nissenbaum 2000:41, (17) for a different judgment).

- (61) [Which picture of himself₁]₂ did [every boy₁ who saw *pg*₂] say Mary liked ____₂?

These examples show that reconstruction into parasitic gaps with internally complex fillers is in principle possible, as predicted by shared antecedent approaches and contrary to the expectations of separate antecedent approaches. What remains to be explained is why parasitic gaps that follow their licensing gaps block anaphor binding reconstruction (e.g. (56b) and (60b)). I leave this as a task for future research, though see Levine and Hukari 2006:49–50 and Bruening and Al Khalaf 2017 for preliminary discussion of a linear component to calculating reconstruction effects in ATB-extraction and parasitic gaps.

3.2 Parasitic gaps display categorial connectivity with their fillers

Let us now turn to a second kind of connectivity, namely categorial connectivity. In a nutshell, I will show—for the first time, to my knowledge—that parasitic gaps obligatorily match their fillers in syntactic category. I contend that this finding supports shared antecedent approaches.

The most widely cited cases of parasitic gaps involve extraction of a DP licensing a DP parasitic gap, as in (62) (I intend ‘DP’ to cover all instances of nominal extraction).

- (62) a. It's [_{DP} Ty] that we can rely on _____{DP} for academic advice
[α _{DP} without necessarily depending on *pg*_{DP} for money].
- b. It was [_{DP} Pepper] that Reuben spoke to _____{DP}
[α _{DP} before Hilary addressed herself to *pg*_{DP}].
- c. It was [_{DP} that door] that Julia peered under _____{DP}
[α _{DP} before sliding a mysterious sealed envelope under *pg*_{DP}].

Additionally, as already observed by Engdahl (1983, 2001) for Swedish and by Levine et al. (2001) for English, extraction of a non-nominal constituent can license a parasitic gap of a matching category. The examples in (63) illustrate with English data: For many speakers, extraction of a PP licenses a PP parasitic gap (see also Steedman 1996:98, n. 41; *pace*, e.g., Cinque 1990, Postal 1993a, 1994, and references cited therein).¹⁹

- (63) a. It's [_{PP} on Ty] that we can rely _____{PP} for academic advice
[α_{PP} without necessarily depending *pg*_{PP} for money].
- b. It was [_{PP} to Pepper] that Reuben spoke _____{PP}
[α_{PP} before Hilary addressed herself *pg*_{PP}].
(modeled after an example in Haik 1987:75, fn. 49, (i))
- c. It was [_{PP} under that door] that Julia peered _____{PP}
[α_{PP} before sliding a mysterious sealed envelope *pg*_{PP}].
(slightly adapted from Murphy 2022:5, (6c))

The extractee can be a l(exically)-selected (e.g. Pesetsky 1991:9–11, Merchant 2019)²⁰ PP argument of the verb ((63a)–(63b)) or a locative PP argument ((63c)), among others.

Crucially, however, the parasitic gap and filler/licensing gap cannot mismatch in syntactic category. Neither does DP extraction license a PP parasitic gap ((64)), nor does PP extraction

¹⁹For evidence that non-nominal parasitic gaps exist in other languages, see Fox 1994:12, (47a) on Hebrew, Manzini 1994:499, (46), (48) and Seguin and Thoms 2025 on Italian, O’Farrill 2023:169–171, app. 5.B on Mexican Spanish, and Ting 2024:7–10 on Mandarin.

²⁰For other work on l-selection, see Haegeman et al. 2014:87, Zyman 2018a,b, 2021:sec. 3, 2022:sects. 6–7, 2023a:sec. 2.3, pp. 26–29, 2024b:esp. supporting information, and Hewett 2023:352, 2024.

license a DP parasitic gap ((65); see Chomsky 1982:55, citing David Pesetsky (p.c.), Hornstein 2001:79, (15), Yoshida et al. 2015:1450, fn. 14, and Hewett 2023:76, fn. 36, 120).²¹

- (64) a. * It's [_{DP} Ty] that we can rely on _____{DP} for academic advice
[α_{PP} without necessarily depending *pg*_{PP} for money].
- b. * It was [_{DP} Pepper] that Reuben spoke to _____{DP}
[α_{PP} before Hilary addressed herself *pg*_{PP}].
- c. * It was [_{DP} that door] that Julia peered under _____{DP}
[α_{PP} before sliding a mysterious sealed envelope *pg*_{PP}].
- (65) a. * It's [_{PP} on Ty] that we can rely _____{PP} for academic advice
[α_{DP} without necessarily depending on *pg*_{DP} for money].
- b. * It was [_{PP} to Pepper] that Reuben spoke _____{PP}
[α_{DP} before Hilary addressed herself to *pg*_{DP}].
- c. * It was [_{PP} under that door] that Julia peered _____{PP}
[α_{DP} before sliding a mysterious sealed envelope under *pg*_{DP}].

Similar facts can be adduced for constituents of other syntactic categories (e.g., Levine et al. 2001:210), though, for reasons of space, I will exclusively discuss DPs and PPs.²²

²¹Chaves and Putnam (2020:45, (105a)) cite an example that is structurally similar to those in (65) as being relatively acceptable, though they attribute its (alleged) acceptability to extragrammatical factors such as memory decay associated with the filler phrase. Notably, I and the speakers I have consulted find their example strongly unacceptable. Thanks to Andy Murphy (p.c.) for bringing this example to my attention.

²²Peter Grishin (p.c.) points out that, in idiolects like his, which allow a pied-piped preposition in an $\bar{A}$ -dependency to surface twice—once at the head of the dependency and once in the extraction site, as in (i) (a phenomenon I refer to henceforth as “P-doubling”; see Nykiel 2010, Radford et al. 2012)—sentences like (65a) are actually quite acceptable.

- (i) % It's [on Ty] that we can rely on for academic advice.

Thus, parasitic gaps mirror non-parasitic gaps in permitting P-doubling in idiolects like Grishin's. Supposing that P-doubling results from multiple spell-out of P in two syntactic positions (on multiple spell-out in the copy theory of movement, see, e.g., Nunes 2004, Landau 2006, Corver and Nunes 2007, van Urk 2018), Grishin reasons that P-doubling in parasitic gap dependencies constitutes yet another species of connectivity effect—one that we can make sense of on a shared antecedent approach where the antecedent occupies all gap positions prior

These facts immediately fall into place on a shared antecedent analysis, where there is a representation of the filler in both gap positions. For instance, in the acceptable sentences in (62a) and (63a), the filler locally satisfies the selectional requirements of both the main clause predicate and the adjunct clause predicate.

- (66) It's $[_{DP} Ty]_1$ that we can rely on $\langle [_{DP} Ty]_1 \rangle$ for academic advice
 $[\langle [_{DP} Ty]_1 \rangle$ without necessarily depending on $\langle [_{DP} Ty]_1 \rangle$ for money].
- (67) It's $[_{PP} on Ty]_1$ that we can rely $\langle [_{PP} on Ty]_1 \rangle$ for academic advice
 $[\langle [_{PP} on Ty]_1 \rangle$ without necessarily depending $\langle [_{PP} on Ty]_1 \rangle$ for money].

In (66), the main clause *P on* and the adjunct clause *P on* both c-select the DP filler *Ty*. In (67), the main clause verb *rely* and the adjunct clause verb *depend(ing)* both l-select a PP argument headed by *on*, which in this case is the *onP* filler. Neither of these predicates can select any other kind of PP or DP complement, as (68) shows.

- (68) We can rely/depend {on/*in/*of/*from/*with/*to/*by/*Ø/...} *Ty* for advice.

When the filler does not locally satisfy the selectional requirements of both predicates, as in (64a) and (65a), the result is ungrammaticality, since selectional features must be checked/satisfied on pain of a crash (see Merchant 2014:1, Zyman 2024b, and Hewett 2023:350, 2025d:6, esp. fn. 8; *pace* Longenbaugh 2019:ch. 5 and Newman 2021).

- (69) * It's $[_{DP} Ty]_1$ that we can rely on $\langle [_{DP} Ty]_1 \rangle$ for academic advice
 $[\langle [_{DP} Ty]_1 \rangle$ without necessarily depending $\langle [_{DP} Ty]_1 \rangle$ for money].
- (70) * It's $[_{PP} on Ty]_1$ that we can rely $\langle [_{PP} on Ty]_1 \rangle$ for academic advice
 $[\langle [_{PP} on Ty]_1 \rangle$ without necessarily depending on $\langle [_{PP} on Ty]_1 \rangle$ for money].

Example (69) is ungrammatical because *depend* l-selects *onP* but its complement is instead a DP. Similarly, example (70) is ungrammatical because *on* c-selects DP but its complement is to movement, thereby feeding multiple spell-out. Importantly, category matching can still be shown to hold in idiolect's like Grishin's, since examples that lack a P-doubling parse, such as (ii), are unacceptable.

- (ii) * It's $[_{PP} on Ty]$ that we can rely _{PP} for academic advice [without necessarily trusting _{DP}].

By contrast, this connectivity is not straightforwardly accounted for by separate antecedent approaches. Consider a null operator analysis of (62a), (63a), (64a), and (65a).

- To explain the grammaticality of (71a) and (71b), *Op* would need to be both categorially and lexically flexible: *Op* would need to be a DP in (71a) so as to be c-selected by *on*, but it would need to be an *onP* (i.e., a PP headed by *on*) in (71b) so as to be l-selected by *depend*.

Second, the idea that *Op* might be categorially flexible, and in particular that it might be able to be a PP, is called into question when we consider other environments where *Op* appears, e.g. *tough*-constructions ((72)). As Browning (1987:254ff.) observes, *Op* cannot be a PP in such contexts; note that this remains true regardless of whether *Op*'s intended antecedent—the subject of the *tough*-predicate—is a DP or a PP.²³

29

(72) * {Ty / On Ty} is difficult [_{CP} *Op*_{PP} to depend _____{PP} for money].

This suggests that true null operators are inherently *nominal*—that is, of category D—echoing one aspect of earlier analyses which took *Op* to be a null pronoun (e.g. Browning 1987, Cinque 1990, Postal 1998, and Munn 2001). If correct, separate antecedent approaches simply cannot account for the existence of acceptable non-nominal parasitic gaps like that in (71b).

Finally, even if PP null operators could somehow be restricted to binding parasitic gaps, a separate antecedent approach would still be incomplete, insofar as the distribution of the alleged PP null operator is not entirely free. Rather, an as-yet unspecified mechanism would need to be invoked to force *Op* and the filler to match in syntactic category and lexical identity to account for the ungrammaticality of (71c) and (71d). I am not aware of any explicit proposal along these lines. Nor do I believe such a mechanism, if devised, would be particularly insightful, since forcing the filler and null operator to be identical in relevant respects (e.g. in binding class, category, lexical identity, and referential index, possibly among others) would be tantamount to adopting a shared antecedent approach.

The upshot is that only shared antecedent approaches offer a straightforward account of
 be interpreted as *By 1999, I will have lived in every city that John has lived in* (see Bresnan and Grimshaw 1978 on similar observations for free relatives).

(i) By 1999, I will have lived in [every city₁ [_{CP} that John has lived ____₁]]. (Larson 1987:240, (2b))

Contrary to what I claim in the main text, then, it might seem reasonable to posit a null operator equivalent to *inP* in (i). Fortunately, there is an alternative account of such missing prepositions that does not require recourse to PP null operators: Larson (1987) argues that the relative-internal P in examples like (i) is elided under antecedent-contained deletion (ACD). Following proposals by May (1985), this ACD is only possible if the entire relative DP (i.e., the relative head and relative clause) undergoes QR above the matrix PP. After QR, the elided PP is no longer contained in the antecedent PP, and there is no infinite regress problem when calculating ellipsis identity at LF. Therefore, I conclude that examples like (i) do not provide evidence for the existence of PP null operators.

categorial (and lexical²⁴) connectivity between parasitic gaps and their fillers. This finding converges with the evidence from binding connectivity laid out in section 3.1.²⁵ What remains is to develop an explicit shared antecedent account of parasitic gaps that derives these connectivity effects. This is the goal of the next section.

4 A new shared antecedent analysis of parasitic gaps

In this section, I develop an analysis of parasitic gaps that combines two very different previous approaches—Nissenbaum 2000 and Nunes 2004—while nonetheless preserving key insights from both. Following Nissenbaum (2000), I analyze *pg*-containing constituents as derived predicates at LF, created by $\bar{A}$ -movement to the edge of the *pg*-containing constituent. This aspect of the analysis derives many of the hallmark properties of parasitic gaps, as I detail in section 4.1. Then, in section 4.2, I lay out an explicit analysis of how $\bar{A}$ -movement works, developing ideas in Hewett 2025a. Specifically, I propose that a constituent α undergoing $\bar{A}$ -movement is encapsulated in a QP shell (Cable 2010a) in its landing site via a layering step of parallel Merge (e.g., Citko 2005). Finally, in section 4.3, I present a new shared antecedent analysis of parasitic gaps and I show how my analysis accounts for the connectivity effects documented in section 3.

²⁴See Merchant (2019) and Hewett (2024) for the proposal that the categorial feature [CAT P] does not exist and that lexical identity with a preposition like *on* may be formally identical to categorial identity with DPs—the former being [CAT *on*] and the latter being [CAT D] (see Zyman 2024b:12, fn. 20 for additional discussion).

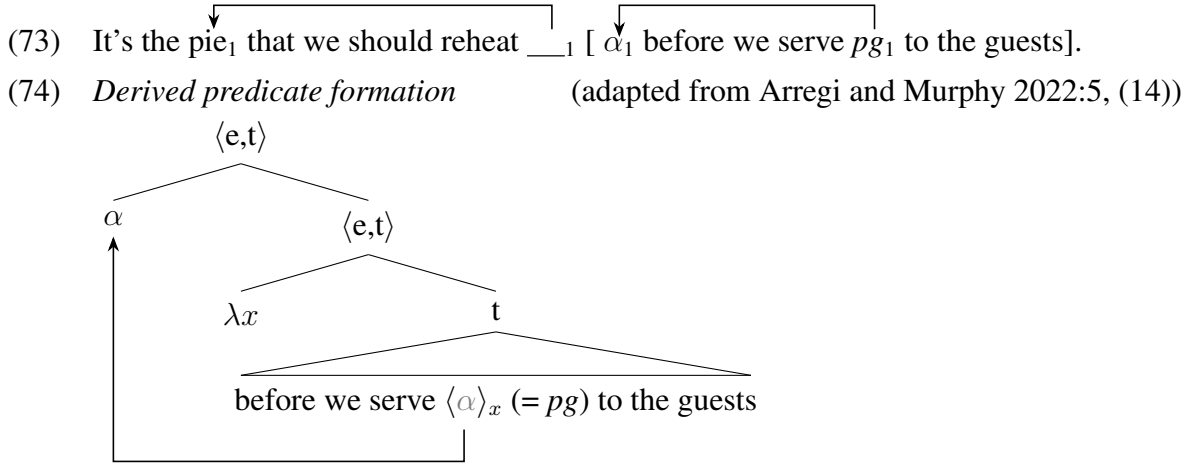
²⁵And note that we can combine the two connectivity tests as in (i)–(iii), with predictable results.

- (i) *PP parasitic gap, Condition C/secondary strong crossover effect*
It's [_{PP} on some other guy₁]₂ that we rely ____₂ for financial advice [without him_{3/*1} depending ____₂ for money].
- (ii) *PP parasitic gap, Condition B effect*
It's [_{PP} on him₁]₂ that we rely ____₂ for financial advice [without him_{3/*1} depending ____₂ for money].
- (iii) *PP parasitic gap, Condition A effect*
It's [_{PP} on himself₁]₂ that he₁ relies ____₂ for food [without PRO₁ also depending ____₂ for money].

4.1 On the distribution and interpretation of parasitic gaps

The following brief primer on Nissenbaum’s analysis draws heavily from Arregi and Murphy (2022:5–9).

Adopting the idea that parasitic gaps involve $\bar{A}$ -movement to the edge of the maximal island containing them (see section 2), Nissenbaum 2000 elegantly accounts for the distribution and interpretation of parasitic gaps via rules of semantic composition at LF. Nissenbaum argues that $\bar{A}$ -movement to the edge of the *pg*-containing island triggers a rule of Predicate Abstraction (e.g. Heim and Kratzer 1998:186, (4)), λ -binding the trace qua variable left by $\bar{A}$ -movement. This creates a *derived predicate* of a higher type, for example $\langle e, t \rangle$. On this analysis, $\bar{A}$ -movement of α to the edge of the *pg*-containing island in (73) creates the derived predicate in (74).

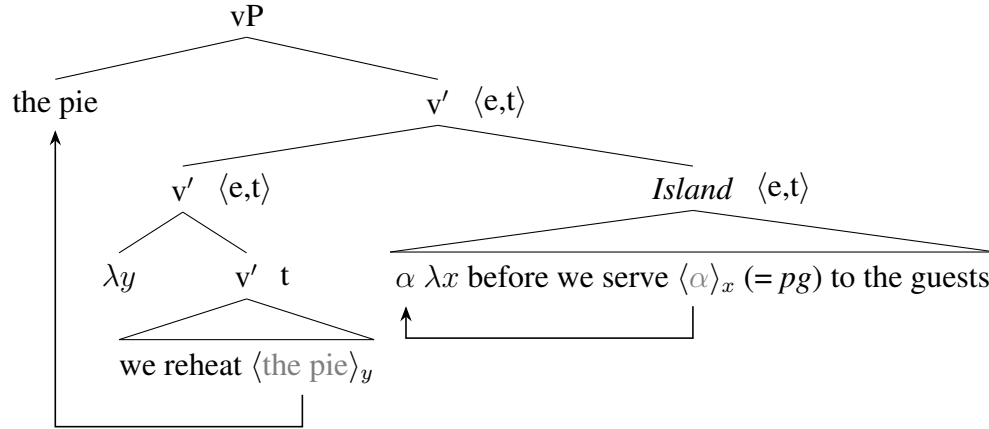


Importantly, for the *pg*-containing constituent (i.e. the mother of α ’s landing site) to be interpreted as a derived predicate, α must be semantically vacuous. On a separate antecedent analysis of parasitic gaps where α is a null operator, this is typically achieved by stipulating that null operators are not interpreted (or, equivalently, are interpreted as an identity function; see Arregi and Murphy 2022:5). Given my arguments against a separate antecedent treatment of parasitic gaps in section 3, I submit instead that intermediate representations in $\bar{A}$ -movement chains can be freely *neglected* (Sportiche 2016) at LF, up to recoverability. Thus, when the

representation of α at the edge of the island is neglected, as it is in (74), the island will remain an open, derived predicate.

Now, in order for the derived, *pg*-containing predicate in (74) to be interpreted correctly vis-à-vis the filler, Nissenbaum argues that it must be sister to *another* derived predicate of the same type at LF. Successive-cyclic $\bar{A}$ -movement of the filler to the edge of vP in the main clause will trigger λ -abstraction and create the requisite derived predicate for the *pg*-containing constituent to attach to, as shown in (75). The two derived predicates semantically compose via a rule of Predicate Modification (Heim and Kratzer 1998:65), yielding a derived predicate that is sister to the intermediate representation of the filler in Spec,vP.

- (75) *Nissenbaum's licensing configuration for parasitic gaps* (adapted from Arregi and Murphy 2022:6, (15))



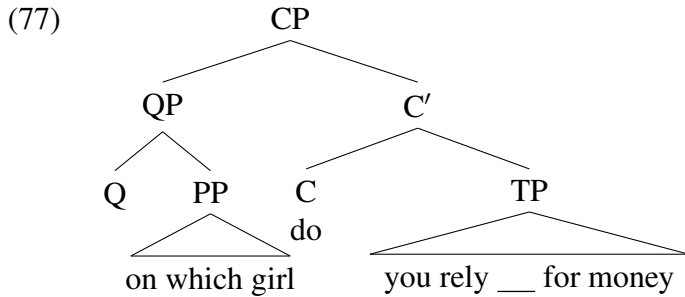
Importantly, on this analysis, the *pg*-containing constituent must be sister to the λ -abstract created by $\bar{A}$ -movement of the filler; if it were to attach any higher or any lower, Predicate Modification would not be possible. While Nissenbaum (1998, 1999:512ff.) derives the phrase marker in (75) by countercyclically adjoining the *pg*-containing constituent immediately below the landing site of filler in the main clause, I will instead assume that λ -abstraction can apply freely (and possibly countercyclically), up to LF interpretability (for an earlier proposal in this vein, see Hewett 2023:400). Thus, the *pg*-containing constituent may (cyclically) adjoin to v' prior to movement of the filler in the main clause. λ -abstraction will then be forced to affect the adjunction site to generate a well-formed expression.

4.2 How $\bar{A}$ -movement works

The next component of my account is an analysis of $\bar{A}$ -movement. As I discussed in section 2, parasitic gaps are created by $\bar{A}$ -movement (and are necessarily licensed by $\bar{A}$ -movement, see section 5). Furthermore, as I showed in section 3.2, the $\bar{A}$ -extraction that licenses a parasitic gap can involve (PP) pied-piping. In order to derive PP pied-piping, I will pursue a variant of Cable’s (2010b) Q-based analysis of $\bar{A}$ -movement, following proposals in Hewett 2025a.

Building on Cable’s foundational work, Hewett argues that, in an $\bar{A}$ -dependency like the *wh*-question in (76), the head of the dependency, which occupies Spec,CP, is not the PP *on which girl*, but is instead a QP whose Q head (which is null in English, by hypothesis) selects that PP as its complement, as in (77).

(76) [CP On which girl₁ do you rely ___₁ for money]?



On this analysis, Q(P)—not the DP *wh*-phrase—bears the feature(s) that the $\bar{A}$ -probe on C (not shown in (77)) is looking for. When Q’s sister is a constituent properly containing the DP *wh*-phrase, like the PP in (77), we derive pied-piping, eliminating the need for a special “feature percolation” mechanism (on which see the references in Heck 2008:6–7). Following Cable (2010b:ch. 6), I adopt a Q-based analysis for *all* $\bar{A}$ -dependencies, including clefts and topicalization, not just *wh*-questions.²⁶

²⁶To derive differences between $\bar{A}$ -dependencies (e.g. selective opacity effects, see Keine 2019, 2020), it may be necessary to distinguish “flavors” of Q(P), one for each $\bar{A}$ -dependency type (e.g. Q_[wh], Q_[rel], Q_[top], Q_[foc], etc.). See Evile and Pesetsky 2023 for an analysis of relative clauses using the Q-like element R.

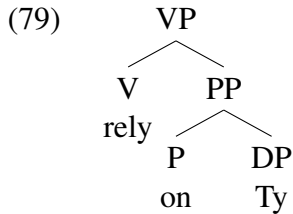
Unlike Cable, however, Hewett contends that QP is not merged in a thematic position; rather, QP is layered on in derived ($\bar{A}$ -)positions only. In this respect, Hewett builds on ideas by Johnson (2012), O’Brien (2017), Poole (2017), Safir (2019), Thoms (2019, 2023), Thoms and Heycock (2022), and Colley (n.d.), among others. Hewett derives such layering via parallel (or sideways) Merge (see Citko 2005 and Nunes 2004, respectively).

Consider the cleft sentence in (78). In order to derive connectivity between the cleft pivot and the gap site (see sections 3–4), I adopt a direct movement analysis of cleft pivots: The pivot moves from the CP-internal gap site to its surface position (see, e.g., Merchant 1998:sec. 5 and Zyman 2024a:9, contra Gundel 1977:sec. 2, among others). For explicitness, I assume that the pivot surfaces in Spec,CP, though no part of the analysis hinges on this assumption.

(78) It’s [_{CP} [on Ty]_I] that we can rely _____I].

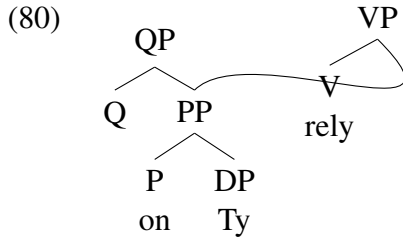
In Pinkham and Hankamer’s (1975) terms, (78) is a *shallow cleft*.

Building from the bottom up, V *rely* externally merges with the I-selected PP argument *on Ty* and projects, forming VP:

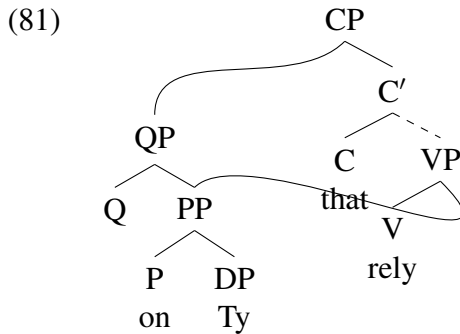


Next, PP undergoes parallel Merge with Q to form QP.²⁷ Following Hewett 2025a, I use *multidominance* to represent structures in which a single syntactic object occupies more than one structural position, as a result of internal or parallel Merge (for more on multidominance, see the references in Citko 2011:44–46 and Murphy 2024:2, among others).

²⁷The exact timing of this step is orthogonal to our discussion: Parallel Merge of Q and PP could be postponed until more of the clause has been built up without affecting the analysis.



Setting aside potential clause-internal intermediate landing sites, the rest of the clausal spine is then built up to C' . At this point, we want to merge QP with C' to effect $\bar{A}$ -movement, as shown in (81).²⁸



But traditional definitions of internal Merge would preclude Merge of QP and C' , since C' does not irreflexively contain QP.²⁹ To get around this issue, I propose that we redefine internal Merge as follows:³⁰

²⁸On a feature-driven approach to Merge, this would be triggered by a Merge-triggering $\bar{A}$ -feature on C' ;

for discussion, see Hewett 2023:sec. 3.2, 2025a:esp. sec. 4.3.

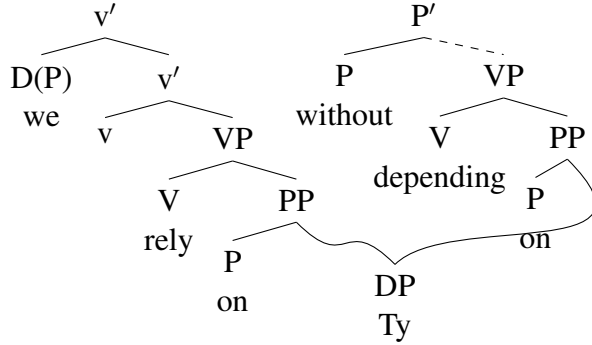
²⁹Some relevant definitions from Collins and Stabler 2016:

- (i) Let A and B be syntactic objects, then B *immediately contains* A iff $A \in B$. (Collins and Stabler 2016:46, Definition 8)
- (ii) Let A and B be syntactic objects, then B *contains* A iff
 - a. B immediately contains A, or
 - b. for some syntactic object C, B immediately contains C and C contains A.
 (Collins and Stabler 2016:46, Definition 9)

³⁰Although Hewett (2025a:22–23, esp. (60)) characterizes Merge of QP and C' in an example identical in relevant respects to (81) as parallel Merge in prose, he derives that Merge with the same feature type that triggers internal Merge, namely “ $\triangleleft$ ” (read: “arrow”) features. The redefinition in (82) abandons any putative distinction between (81) and more canonical instances of movement by treating them all as internal Merge. Note that more

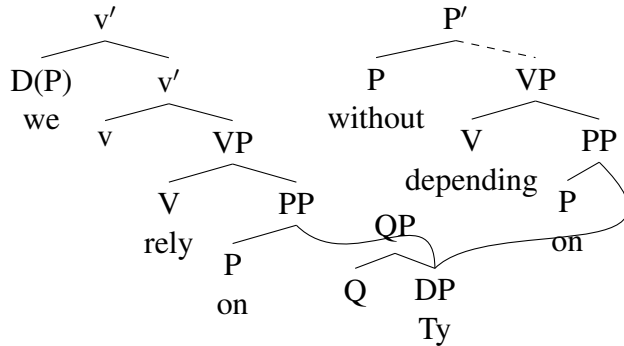
complement of both *on*'s) via external/parallel Merge, yielding a sharing construction ((84)). For explicitness, I analyze the island as a PP headed by *without*. Moreover, I use dashed branches to abbreviate irrelevant portions of the trees.

(84) *Derivation of (83), step 1: Build two clauses in parallel, sharing DP*



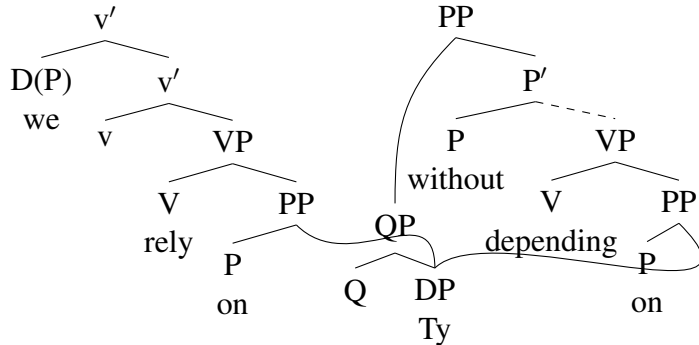
Next, Q is layered onto the DP filler via parallel Merge, forming QP.

(85) *Derivation of (83), step 2: Parallel Merge of DP with Q, forming QP*



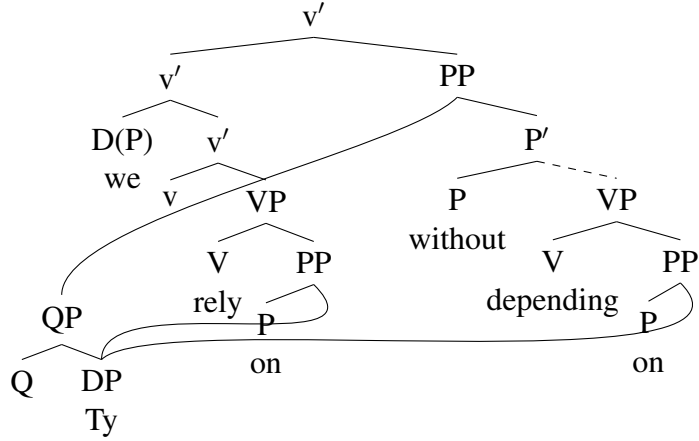
This QP then undergoes internal Merge to the edge of the adjunct, in compliance with the definition of internal Merge in (82).

(86) *Derivation of (83), step 3: Internal Merge of QP with P'*



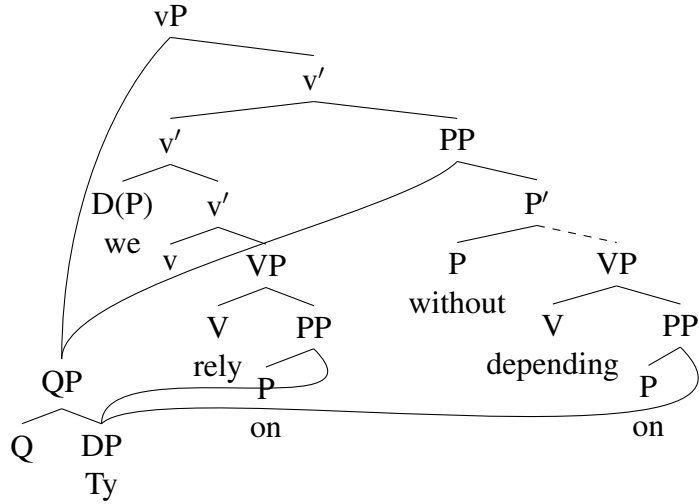
With the adjunct island now completed, it adjoins to v' .

(87) *Derivation of (83), step 4: Adjunction of PP island to v'*



Finally, QP is internally merged in Spec,vP , above the adjunction site of the PP island.

(88) *Derivation of (83), step 5: Internal Merge of QP with v'*



There are at least two questions which need answering at this juncture: (i) How can QP move to Spec,vP , given that QP is irreflexively contained inside an adjunct island (i.e., why isn't (88) unacceptable due to illicit extraction out of an adjunct island)? And (ii) How is the structure in (88) interpreted? I answer each question in turn.

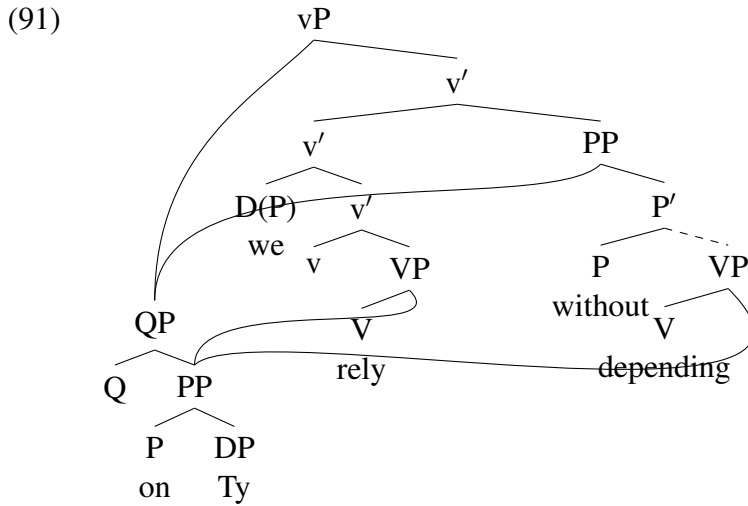
Regarding (i), internal Merge of QP and v' in (88) is permitted by the definition of internal Merge in (82). Taking v' as α and QP as β , (82) states that internal Merge of v' and QP is defined if and only if every constituent γ that QP irreflexively contains that is distinct from the

and Poole (2017) for proposals regarding how to interpret multidominant *wh*-movement structures that could be adapted to what I have proposed here.

The derivation of a PP parasitic gap like that in (90) is identical in relevant respects to the derivation of a DP parasitic gap, the difference being that the former involves sharing a PP in both clauses.

- (90) It's [PP on Ty] that we can rely ____{PP} for academic advice
 [α_{PP} without necessarily depending pg_{PP} for money].

Example (91) illustrates my analysis of (90).



To summarize, I have proposed that parasitic gap constructions are generated via $\bar{A}$ -movement of the filler in the *pg*-containing constituent and in the main clause. $\bar{A}$ -movement involves layering a QP shell over the moving constituent. While I have focused on DP and PP extraction, see, for instance, Levine et al. 2001 on acceptable parasitic gaps of other categories in English. Finally, parasitic gap constructions are interpreted according to Nissenbaum's (2000) analysis—namely, via Predicate Abstraction and Predicate Modification.

4.4 Deriving connectivity in parasitic gap constructions

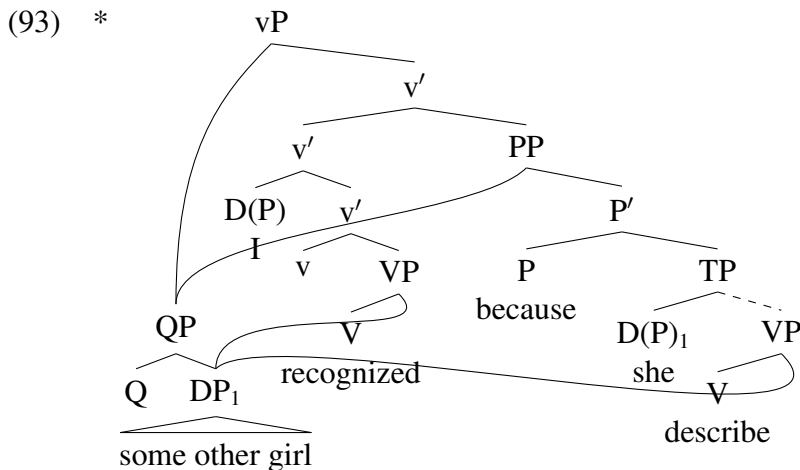
Having laid out my shared-antecedent, layering, multidominant analysis of parasitic gaps, I will now explain how this analysis derives the connectivity effects adduced in section 3. First, I explain binding connectivity in section 4.4.1, and then I account for categorial connectivity in section 4.4.2.

4.4.1 Deriving binding connectivity

Recall from section 3.1 that parasitic gaps match the Binding-Theoretic class of their fillers. Thus, the parasitic gap in (92) (repeated from (43)) displays a Condition C/strong crossover effect because the filler it is coindexed with is a quantificational expression.

- (92) It was some other girl₁ that I recognized ___₁ [_{α_1} because she_{3/*1} taught you to describe *pg*₁ so well].

This connectivity effect follows straightforwardly from structure sharing: Because the filler occupies all gap positions, via parallel Merge, prior to movement, the Binding Conditions will necessarily be evaluated over a representation that includes the DP content of the filler in both gap positions. For instance, in (93), the filler *some other girl* is A-bound by *she* in the adjunct clause, triggering a Condition C effect.



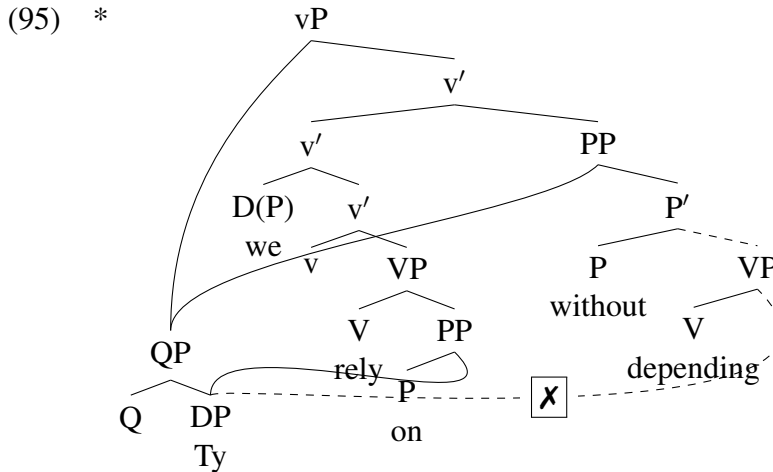
Condition B ((49)) and Condition A ((50)) effects with parasitic gaps can be similarly derived, though, for reasons of space, I do not go through explicit derivations here.

4.4.2 Deriving categorial connectivity

My shared antecedent analysis likewise accounts for connectivity between the filler and parasitic gap in syntactic category (and lexical identity). First, recall the fact that DP extraction does not license a PP parasitic gap.

- (94) (repeated from (64a))
 *It's [_{DP} Ty] that we can rely on _____{DP} for academic advice
 [_{α PP} without necessarily depending *pg*_{PP} for money].

As discussed above, this follows straightforwardly as a violation of selectional requirements in the *pg*-containing constituent. Specifically, what is shared between the two clauses in (94) is just the DP *Ty*, as shown in (95). Yet, this DP cannot satisfy the l-selectional requirement of the adjunct-internal V *depend*, which is looking to merge with a PP argument headed by *on*.



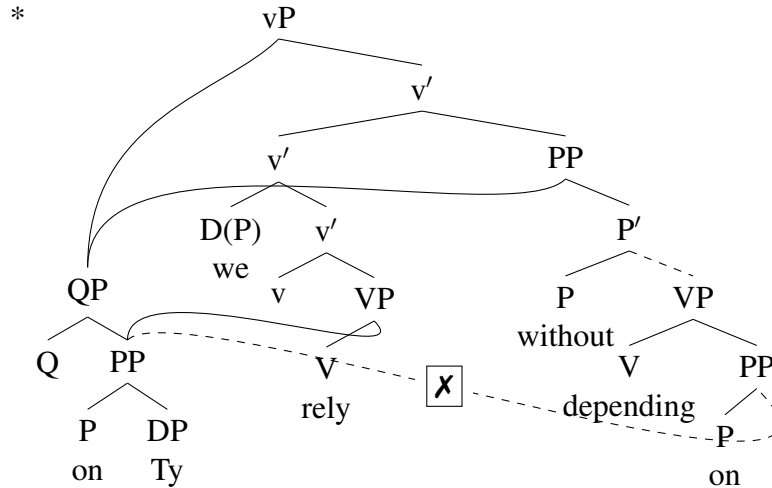
We also saw in section 3.2 that PP extraction does not license a DP parasitic gap.

- (96) (repeated from (65a))
 *It's [_{PP} on Ty] that we can rely _____{PP} for academic advice
 [_{α DP} without necessarily depending on *pg*_{DP} for money].

There are at least two possible derivations of (96) that need to be considered and, ultimately,

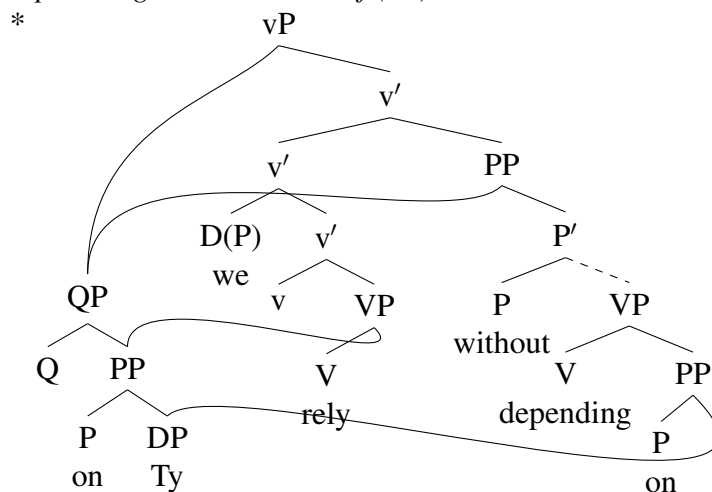
ruled out. The first involves sharing of the PP portion of the filler in both clauses, as in (97). As with (95), such a derivation will never be generated on a shared antecedent approach due to a violation of selectional requirements in the *pg*-containing constituent: [_P *on*] in the adjunct c-selects a DP complement, not a PP/*on*P.

(97) *A failed derivation of (96) involving PP sharing*



This naturally leads us to consider an alternative derivation where the ‘correct’ constituent is merged in each clause. Specifically, what rules out a pathological derivation of (96) along the lines of (100), where [_{DP} *Ty*] is merged in the parasitic gap position and [_{PP} *on Ty*] is merged in the licensing gap position, prior to $\bar{A}$ -movement?

(98) *A pathological derivation of (96)*



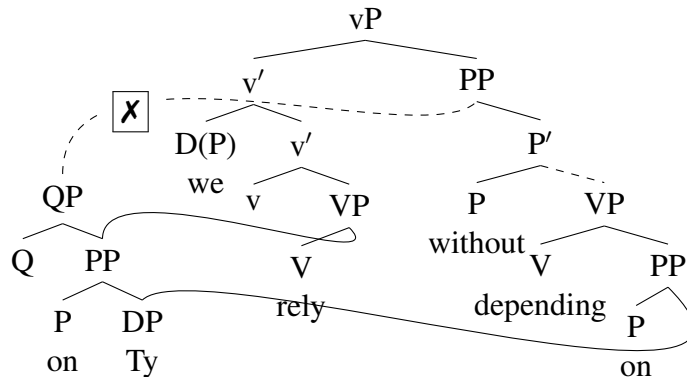
It so happens that the specific definition of internal Merge I proposed in (82), repeated below as (99), excludes such a derivation.

- (99) Internal Merge of α and β is defined iff for every constituent γ such that
- (i) β irreflexively contains γ , and
 - (ii) the label of γ is distinct from the label of β ,
- the head that α is a projection of asymmetrically c-commands γ .

As I noted at the end of section 4.2, this definition of internal Merge enforces an upper limit on layering: Only one head can be layered onto a moving constituent at a time. This is because, for a given step of internal Merge, the attracting head (i.e., α in (99)) must asymmetrically c-command *every constituent* γ contained in β that is distinct (in sense of Bare Phrase Structure; Chomsky 1995) from β .

With this condition in mind, consider once more the pathological derivation in (100), focusing on the two steps of internal Merge of QP. Internal Merge of QP and v' does not run afoul of the condition in (99): v' does asymmetrically c-command every constituent contained in QP that is distinct from Q(P). Internal Merge of QP and P' , however, is prohibited by (99), since P *without* does not asymmetrically c-command the $P(P)$ contained inside QP. In other words, the derivation in (100) crashes due to illegal layering of more than one constituent (i.e., PP and QP) in the single step of internal Merge to the edge of the *pg*-containing island. This is shown in the following tree:

- (100) *A failed derivation of (96) involving illegal multiple simultaneous layering*
*



By ruling out the pathological derivation in (97), we have successfully derived all the categorial connectivity effects documented in section 3.2.

In summary, I have proposed a shared antecedent analysis of parasitic gaps that makes use of a constrained form of *layering*—combining parallel Merge and internal Merge—in the derivation of $\bar{A}$ -movement. This analysis derives connectivity between parasitic gaps and their fillers/licensing gaps as an identity effect under displacement. Importantly, this is a success arguably not shared by separate antecedent approaches to parasitic gaps, which have no straightforward account of the aforementioned connectivity effects. Before concluding, I will note one more potential advantage afforded by my analysis of parasitic gaps.

5 Layering and the A-/ $\bar{A}$ -distinction

A well known, though arguably not well understood, property of parasitic gaps is that they are only licensed by $\bar{A}$ -movement (Engdahl 1983). Thus, $\bar{A}$ -movement of the cleft pivot in (101) can license a parasitic gap in the clause-final vP adjunct headed by *before*.

(101) $\downarrow$ $\bar{A}$ It's the apple pie₁ that we should reheat ₁ [before we serve *pg*₁ to the guests].

However, A-movement, for instance passive A-movement of the internal argument of a transitive verb to the surface subject position in (102), does not license a parasitic gap.

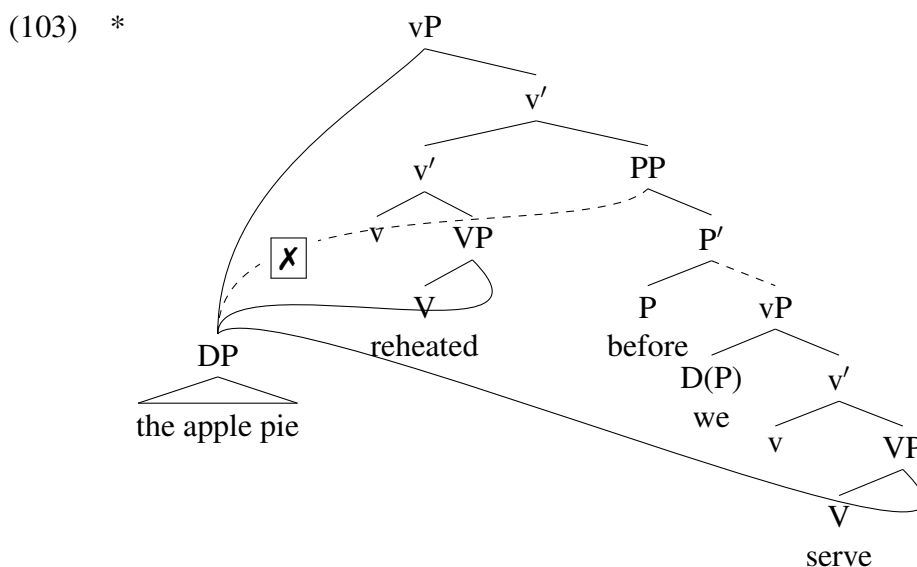
(102) $\downarrow$ A *The apple pie₁ should be reheated ₁ [before we serve *pg*₁ to the guests].

Despite recent attempts to derive this asymmetry from independent factors (see, e.g., van Urk 2017 for an account based on the semantic types of the predicates created by A- and $\bar{A}$ -movement), no analytical consensus has emerged. Given this uncertain state of affairs, it is noteworthy that the analysis of parasitic gaps I developed in previous sections actually offers a novel way to account for the A-/ $\bar{A}$ -distinction in parasitic gap licensing. Specifically, the requirement that parasitic gaps be licensed by $\bar{A}$ -movement can be understood as another

manifestation of categorial connectivity under movement.³¹

To see how, recall the hypothesis from section 4.2 that $\bar{A}$ -dependencies are headed by QPs (Cable 2010b). Furthermore, suppose that A-dependencies are *not* headed by QPs (Safir 2019); instead, they are typically headed by DPs (Hewett 2025a). Then, granting the fact that parasitic gaps are, by definition, gaps created by $\bar{A}$ -movement (section 2), only QP movement should be capable of licensing a parasitic gap (qua gap created by QP movement).

Consider the derivation of the vP in (102) shown in (103).



The filler is a DP that has undergone A-movement to subject position in the main clause. Suppose, for the sake of argument, that such A-movement passes through Spec,vP (or a similar clause-medial specifier position; see, e.g., Sauerland 2003, van Urk 2017:esp. 537, 541, Hewett 2025c,d). Because the filler is a DP and not a QP, what moves inside the PP adjunct island must also be a DP. But without QP movement to the edge of the island, no parasitic gap can be created. Indeed, if DP movement to the edge of the island behaves like A-movement and if A-movement is generally constrained by DP-relativized minimality (e.g. Hewett 2025a:sec. 3.2.1 and references therein), then we predict (102) to be ungrammatical since A-movement of DP to the edge of the adjunct violates (feature-based) Relativized

³¹For a related, but distinct, idea, see Safir 2019:sec. 4.3.

Minimality. Thus, parasitic gaps cannot be licensed by A-movement because A-movement fillers bear the wrong syntactic category; the requirement that parasitic gaps be licensed by $\bar{A}$ -movement reduces to a category matching effect.³²

6 Conclusion

The core novel empirical contribution of this paper is the discovery that parasitic gaps display under-recognized connectivity with their fillers in binding theoretic class and syntactic category/lexical identity. Shared antecedent analyses of parasitic gaps offer the most straight-

³²Of course, this analysis begs the question: Can A-movement of a DP nonetheless license a ‘parasitic’ DP gap created by A-movement (i.e., one that obeys minimality constraints on A-movement)? Initial results suggest that it can, though I must leave a fuller investigation into this matter for future work. Pylkkänen (2008:35–40) observes that secondary depictive predicates (henceforth: depictives) are licensed by A-movement, for instance in verbal pseudopassives like (i) (see also Bruening 2011:388, fn. 13 and van Urk 2017:536–537); note that I use subscripted indices to indicate which argument the depictive is predicated of.

- (i) This bed₁ was slept in ____₁ [Depictive still wet₁].

As a control, example (ii) illustrates that depictives cannot be predicated of an in situ complement of P.

- (ii) The dog₁ slept in this bed₂ [Depictive still wet_{1/*2}].

Crucially, depictive phrases can be (or contain) (pseudo)passive participles, as shown by the following examples:

- (iii) I can’t believe this tent₁ was slept in ____₁ [Depictive {already broken into ____₁ / rained on ____₁}].
 (iv) This bed₁ was slept in ____₁ [Depictive covered ____₁ with dust].

If (pseudo)passive participles are formed by predicate-creating A-movement (for arguments to that effect, see Bruening 2014:esp. 385–387, Hewett 2025c), then (iii)–(iv) reveal that A-movement of a DP can license a ‘parasitic’ DP gap left by A-movement. A schematic, shared antecedent analysis of one of the examples in (iii) is given in (v).

- (v) ... [DP this tent]₁ was slept in <[DP this tent]₁> [<[DP this tent]₁> rained on <[DP this tent]₁>].
-

forward account of this connectivity, contrary to the current consensus, which maintains a separate antecedent approach. My findings therefore support a distinction between parasitic gaps (and their antecedents) and trull null operators, as found in *tough*-movement and (certain) *that*-relatives (*pace*, e.g., Chomsky 1986, Browning 1987, Lasnik and Stowell 1991, Nissenbaum 2000, Hornstein 2001). The shared antecedent analysis I developed preserves and synthesizes key insights of quite different earlier approaches (in particular, Nissenbaum 2000 and Nunes 2004). Crucially, in order to explain category matching effects, I proposed that internal Merge must limit layering to applying, maximally, one head at a time. Such a condition is attractive not only because it derives the facts, but also because it adds restrictiveness to a grammar with both parallel and internal Merge (contrast, e.g., Thoms 2019, Thoms and Heycock 2022, where layering is unrestricted/unbounded). It remains to be seen whether independent support can be found for this restriction.

Looking beyond parasitic gaps, previous work has shown both of the connectivity effects documented here to likewise be attested in ATB-movement; for connectivity in binding theoretic class, see Bruening and Al Khalaf 2017 on English, and for connectivity in syntactic category, see Georgi et al. 2025 on Greek.³³ The evidence from connectivity therefore provisionally supports a symmetric or shared antecedent approach to ATB-movement (Ross 1967, Williams 1978, Citko 2005, Bruening and Al Khalaf 2017, among others; *pace*, e.g., Franks

³³Category matching between the filler and all gaps is also obligatory in English ATB-movement, as (i)–(ii) show.

- (i) a. It was [_{DP} that door] that Ty [peered under _____{DP}] and [carefully opened _____{DP}].
- b. It was [_{PP} under that door] that Ty [peered _____{PP}] and [slid a secret sealed envelope _____{PP}].
- c. *It was [_{DP} that door] that Ty [peered under _____{DP}] and [slid a sealed secret envelope _____{PP}].
- d. *It was [_{PP} under that door] that Ty [peered _____{PP}] and [carefully opened _____{DP}].
- (ii) a. It's [_{DP} Ty] that we can [rely on _____{DP} for advice] and [depend on _____{DP} for money].
- b. It's [_{PP} on Ty] that we can [rely _____{PP} for advice] and [depend _____{PP} for money].
- c. *It's [_{DP} Ty] that we can [rely on _____{DP} for advice] and [depend _____{PP} for money].
- d. *It's [_{PP} on Ty] that we can [rely _____{PP} for advice] and [depend on _____{DP} for money].

1992, Munn 1992, 1993, Ha 2008, Salzmann 2012a,b). If this conclusion is correct, it points to a broader analytical unification of multiple gap dependencies.

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